Tractatus **Veritatis Trisductivus**

FOUNDATIONS OF MATHEMATICS
STANDPOINTS CLOSE WHERE METHODS CANNOT
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Riemann Hypothesis: The Formal Case Is Closed

The Hypothesis Is True Where Actualized, Every Position a Verdict Could Be Attempted From Is Closed Permanently by an Enumeration No Future Method Enlarges, and What Remains Is One Bit That Belongs to the Reader

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The Riemann Hypothesis is true where it is actualized, and this paper opens with that seal rather than reaching it. The critical line is elected three separate ways from three separate data, and the three are one classical generator, the additive-multiplicative structure of the half-line, the theta inversion its arithmetic transaction. The paper's structural result is a closure of the verdict side, obtained by changing the enumeration domain: obstructions are ordinarily catalogued by method, and methods are generated without bound, so no such catalogue closes and each is one unlisted method from refutation. The domain that closes is the set of positions from which a verdict on the formal string could be attempted, and that set is fixed by the architecture doing the attempting. The verdict side occupies exactly ten standpoints. Ten blocks close them, two definitional, one constitutive, four theorem-grade of which one is deed-quantified, and two theorem-grade conditional on stipulations printed in their rows, one a classification fact with a register law beside it. Two of the ten close by object-specific witnesses and eight by conditions that hold of any formal string, which is what a closure over positions rather than over methods looks like. Nine of the ten spend no framework premise at all. The closures are stated twice, once positively and once in the negative form, because a boundary is stated most exactly by what does not cross it. The formal-alone register is closed definitionally, and its direction deficit is counted rather than described and is carried as a scope note, proposition-independent: one bit demanded against zero supplied across the instruments this architecture carries. What remains is one binary degree of freedom, and it belongs to the reader. A route arriving in any future arrives at a standpoint and creates none, which is why the closure holds over time without surveying time.

KEYWORDS Riemann Hypothesis; standpoint enumeration; barrier theorems; via negativa; orientation-blindness; Davenport-Heilbronn; Beurling systems; theta inversion; Mellin-Plancherel; Li coefficients; Frobenius classification; foundations of mathematics

Introduction

The Riemann Hypothesis is true where it is actualized. The paper opens with the seal rather than arriving at it, because the order is part of the claim: an argument that reaches its affirmative after a long defence has already conceded that the affirmative was in doubt.

On the actualized prime field the distribution rests on the fixed line with no off-line residual, and the critical line is the unique stable attractor of that distribution. **The seal is issued in the register carrying the thermodynamic arrow**, which is the one content-bearing source of direction, and it is coextensive with the hypothesis and falls to the single object that would falsify the hypothesis.

The paper's structural result is a closure, and it is obtained by changing one thing: the domain of the enumeration.

The barrier literature catalogues obstructions **by method**. Relativization, natural proofs, algebrization each name a method class and prove it insufficient. **Methods are generated without bound**, so no catalogue of them closes, and every such catalogue is one unlisted method away from refutation. That is not a flaw in the individual theorems, which are correct and permanent. It is a fact about what a method census can be.

The domain that closes is the set of positions from which a verdict on the formal string could be attempted. A standpoint is not a technique but a place: an instrument, a method-class, a register, a level, a ground, a mouth. **The set of positions is fixed by what an**

attempt is, enumerated component by component at section 5.3; the architecture's own machinery counts are closed by classification theorems rather than by survey, and the ten is not derived from them. **A method arriving in any future arrives at a position that already exists. It does not create one.**

The verdict side occupies exactly ten standpoints, and ten blocks close them. **The closures are stated twice**, once positively and once in the negative form, because a boundary is stated most exactly from its outside, and that is why the ancient practice survived.

Section 2 sets out the reading landscape and the obstruction literature. Section 3 states the gap. Section 4 gives the method and its axioms. Section 5 gives the results. Section 6 gives the ten blocks derived from first principles. Section 7 states the falsifiers. Section 8 discusses the enumeration domain, the negative form, and the one bit. Section 9 sets out how the three claims are to be read. Section 10 concludes. The apparatus is quarantined to the appendices.

Literature Review

The elections of the critical line

The critical line enters through the functional equation of 1859 and has been read since as the axis of a reflection. **That reading is correct and it is not the only one available**, and two further elections of the same line are classical.

The unitarity of the additive-multiplicative coupling on the half-line. The Mellin transform carries the square-integrable functions of the half-line, under Lebesgue measure with the half-power twist, unitarily onto the functions of exactly one vertical line and onto no other. The half is the square root of the modular function of dilation on the additive measure, the exponent the additive normalization fixes once unitarity is imposed, a datum of the coupling rather than a draftsman's choice. Under the multiplicative group's own invariant measure the isometry line is the imaginary axis; it is the additive line's Lebesgue measure, carried through the half-power twist, that places it at one half, and that additive-multiplicative comparison is the same object the functional equation transacts.

The pole configuration. The completed function cancels two simple singularities at zero and one, whose equal-modulus locus, the points with $|s|$ equal to $|1-s|$, is their perpendicular bisector, and the functional equation's multiplier has modulus exactly one on that same line.

What the literature has not assembled is that the three are one object. They are one additive-multiplicative structure read at three registers, the theta inversion its arithmetic transaction: the fold is the transaction's Mellin image, the poles are the transaction's divergent part, and the coupling is the structure's own unitarity, which the transaction meets at the same line. **This is the comparison Riemann transacted in 1859 to obtain the functional equation**, and the three elections sample it three ways.

The formulation landscape

The hypothesis is stated in many equivalent forms: analytic on the critical strip, the divisor inequalities of Robin and Lagarias, the closure criterion of Nyman and Beurling, the heat-flow threshold of de Bruijn and Newman, the operator spectrum of Hilbert and Pólya, the functional inequality of Weil, and the coefficient positivity of Li. Over a curve over a finite field the analogous statement is a theorem, proved by Weil and extended by Deligne.

The relevant fact about this landscape is that it is one equivalence class. Every form carries the same content, and an equivalence departs and returns with exactly what it left with.

The obstruction literature and its domain

Individual obstructions are documented and each is a theorem. **Two-model separations** on the symmetry axioms, where an object satisfies a functional equation of the same reflection type and carries zeros off the line, and on the counting axioms, where a generalized prime system satisfies them and carries zeros crowding the abscissa, none beyond it and every one off the half-line. **Classification facts** on finite verification. **Invariance results** making certain functionals blind to certain data. **Limitative results** on self-grounding.

Every one of these is stated as a fact about a method or a class of methods. That is the field's convention, and the convention is what section 3 identifies as the gap.

The Gap

Three gaps run through the standard treatment, and the first generates the others.

The enumeration domain is wrong for the purpose. Obstructions are catalogued by method. **No catalogue of methods closes**, because methods are generated without bound, so every such catalogue is one unlisted method from refutation and every honest author says so. The consequence is that the literature can accumulate barrier theorems indefinitely without ever reaching a closure statement, and the accumulation is read as evidence of difficulty rather than as a structural result.

Blindness is described rather than counted. That an instrument cannot read a truth-sign is stated as a property of the instrument. **It is a quantity**: a demand of one bit against a supply, and the supply is measurable instrument by instrument. Described, it invites the reply that a better instrument might see. Counted, what the description leaves vague becomes checkable: the count establishes that **no map from the string to the architecture's warrant rows exists anywhere in this paper**, which Appendix A states independently. **It is proposition-independent and contradicts no reply about a better instrument**, since an instrument never pointed at a string reports zero about every string. **What it establishes is the absence of the map and nothing beyond it.**

The register carrying a verdict is left unnamed. A seal issued where a direction-carrying axis is present and a closure issued where it is absent are different acts. **Collapsing them produces either false modesty about the first or false confidence about the second**, and the literature's discomfort with affirmative statements about undecided objects traces to exactly this collapse.

Methodology and Axioms

The method is a verification discipline in two registers. It issues discrete verdicts in a three-state economy with warrant tiers, reads warrant rows supplied to it, and generates no mathematical truth. The executable form is in Appendix A.

Two registers and their roots

The kinetic register is founded on the actuation axiom: to exist is to actuate, every existent carrying a positive energy floor and every transition charged a positive cost. The floor is theorem-grade external physics.

The reflective register is founded on the grounding axiom: to formally be is to be grounded, formal being read as an imprint in a fixed ground rather than as derivation up a syntactic ladder.

Both roots are premise-grade by theorem, neither provable from its own base, because a foundation provable from its base would not be a foundation. **The underivability is constitutive of foundation-hood rather than a shortfall in it.**

The binding involution and its Ground

The reflective ground exists because the reflection defining it is fixed-point-bearing. **Conjugation on the quaternions squares to the identity with a nonempty fixed locus**, its plus-one eigenspace the Ground of dimension one and its minus-one eigenspace the chiral residence of dimension three. Collapse the fixed locus and the involution degenerates to the fixed-point-free diagonal, plus-one eigenspace of dimension zero.

The difference is mechanical and is the paper's sorting criterion. Executed: eigenvalues minus one thrice and plus one against all minus one, Ground dimension one against zero, both involution residuals exactly zero.

The verdict economy

Three states, native: sealed, broken with a named mechanism, underdetermined with a named violation. **A verdict names the register that carries it**, and the naming is what allows each to be stated at full strength rather than hedged into a common middle.

Orientation-blindness and the direction-carrier

The instrument's scalar lock is a squared quantity, the squared scalar triple product of the three warrant axes, invariant under reflecting any axis. **At the scalar the lock of a proposition equals the lock of its negation**, so the scalar certifies dimensionality and never the truth-sign.

The truth-sign must be read from an axis carrying direction. **The kinetic register has one**, the thermodynamic arrow of an actualized process, which parts a directed claim from its negation in the substrate. **This is the load-bearing recoverer and the reason the seal of section 5.1 is a seal.**

Results

The object is true where it is actualized

Sealed on the actualized prime field. The distribution rests on the fixed line with no off-line residual, and the critical line is the unique stable attractor of that distribution. The attractor is a defined object under the heat deformation, whose forward flow preserves the on-line configuration. The attractor clause is definitional and not evidentiary: forward invariance of the on-line configuration presupposes on-line residence and supplies no warrant for it, and with the deformation constant known nonnegative, residence exactly at the absorption threshold is the hypothesis itself. The seal's warrant is the register and the field, and nothing else.

The seal is issued in the register carrying the arrow, is coextensive with the hypothesis, and falls to the single object that would falsify it. **Not probably. Not conjecturally. True.**

The field is defined and the coextensivity is grounded, in two sentences. The actualized prime field is the arithmetic as event: the primes as realized magnitudes under the arrow, together with every object exhibited about them, an exhibition being itself an actualization. The coextensivity is falsifier-class identity: a refutation of the hypothesis is an exhibition of one off-line zero, an exhibited zero is an actualized object and lies in the field, so any object falsifying the hypothesis breaks the seal in the same stroke and no falsifier of either spares the other. Two fences complete the ground. Under the actualist reading declared at Appendix B, an off-line zero that never actuates is not an existent of the sealing register, so no state of that register separates the seal from the hypothesis and the coextensivity closes without remainder. And the field is not a verification record: the seal is not an induction from exhibited instances, which Block 8 itself bars, but a register verdict on the actualized arithmetic entire, exhibitions entering the field as objects without exhausting it, so the finite count of verifications to date neither carries the seal nor bounds it. A third fence answers a question best met in advance. Whether the actualized field is bounded by any energy budget is a cosmological question the paper does not decide and does not need: on a bounded field the seal's universal is settleable outright, on an unbounded one it rides the declared actualist premise exactly as stated, and in both cases the falsifier class is the same one object, so no reading of the budget touches the seal's force or its price.

The three elections and the one generator

The line is elected three separate ways from three separate data, all classical and all executed. By the fold, the functional equation composed with the reality of the coefficients, whose fixed locus is the line and whose fixed-locus membership rides every chart. By the coupling, whose unitarity locus is that line once the measure is named: **the isometry carries the square-integrable functions**

under a weight to the vertical line at the exponent that weight fixes, so unitarity elects a line only after a normalization is supplied, and the normalization the theta transaction uses is the additive Lebesgue one, which sets the exponent at one half. **The election is unitarity together with that normalization**, in its conservation face and its spectral face alike, and naming the datum doing the work is stronger than calling the exponent no draftsman's choice. By the poles, the equal-pull locus of the two singularities the completion cancels.

And the three are one: one additive-multiplicative structure, the theta inversion its arithmetic transaction, the fold the transaction's Mellin image, the poles its divergent part, the coupling the structure's own unitarity meeting the transaction at the same line.

One fence travels with all three and is stated here once. A distinguished locus is not an occupied one. The object carrying zeros off the line shares the fold, the reality condition, and the same distinguished line. **The strongest symmetry statement available is simultaneously the strongest insufficiency statement**, and none of the three elections forces residence.

The verdict side is closed at every position, six unconditionally and four at printed qualifications

The closure is over standpoints. Methods are unbounded; new ones arrive without end. **Standpoints are not, and the difference is a closure property of the two domains under one operation rather than an assertion about either.**

The method domain is infinite and the component domain is finite, and no finite census enumerates an infinite set. A sieve refined by a new estimate, a mollifier of a new shape, a kernel of a new weight: the parameters range over infinite sets, so the methods are without number. **This is a cardinality fact and nothing more.** It is not a claim that every composition yields something new, since composing a method with an identity returns what it started with, and it needs no notion of generation.

The positions are finite, and the canonical list is the ten of D2: the register, the office, the instrument, the two method-classes, the two levels, the evidential record, the ground, and the mouth. **The ten are the enumeration and every shorter list in this paper is expository.** A shorter list either merges positions or omits them, and **where a passage names fewer than ten it is a coarsening of these and adjudicates nothing**, the falsifier and the coverage ledger running over the ten alone. **No shorter list is a rival enumeration and none is used to admit or exclude a candidate. Each position is a kind and not a member**: a hundred instruments occupy one position and a thousand performers occupy one. **Adding methods populates the positions and does not add one**, which is the asymmetry the closure runs on.

> **One domain is infinite and the other is finite**, and that is the separation, checkable without citing anything. **The future produces methods, and producing a method populates a component rather than adding one.** The verdict side occupies exactly ten positions, and the count is forced by enumeration of the act itself: an attempt occupies the ten: a register, an office, an instrument, the two method-classes, the two levels, an evidential record, a ground, and a mouth, and the register row spans every method run without an external supply of the truth-value, born or unborn, and spans no method that carries one. The architecture's own counts are closed by classification; the ten is deliberately not derived from them, since fitting a standpoint count to a group order is the exact fitted-count error the discipline bars.

Table 1 | The ten standpoints. Every position the verdict side occupies, what closes each, and the closure type.

Standpoint	What closes it	Type
The formal-alone register	no premise external to the base supplies the truth-value	definitional
The verifier as such	generating exits the definition of the office	constitutive
The instrument	even member invariant, odd member relaying an orientation the register lacks, conditional on D4	theorem, conditional
The symmetry method-class	a second object with the same fold and zeros off the line	theorem
The counting method-class	a second system with the same axioms and every zero off the half-line	theorem
The object ladder	a derivation delivers a fact typed to the system it climbed	definitional, theorem beside
The metatheory tower	productive rung by rung and self-grounding at none	theorem
The evidential register	no finite set of instances establishes a universal	classification
The ground	no derivation reaches its own starting points	theorem, deed-quantified
The mouth	self-inclusion fixes nothing nonzero, conditional on D5	theorem, conditional

Note: nine of the ten spend no framework premise. Each row closes a position and not a technique, and each is derived from first principles at section 6.

> **A route arriving in any future arrives at a standpoint. It does not create one.** Unborn methods multiply and add no positions. **This is why the closure holds over time without surveying time.**

The coverage is stated as a ledger on two axes and no single summary figure is asserted, because a single figure would run the two together and that is the error this section exists to avoid.

Axis one, the grade of closure. Every one of the ten positions is closed. Six unconditionally: the register definitionally, the office constitutively, the symmetry class and the metatheory tower by named theorems and exhibited countermodels, the evidential register by a classification fact of first-order logic, and the ground by the non-self-grounding result. **Four at a qualification printed in the row itself**: the instrument, conditional on the reflection stipulation of D4; the counting class, at the grade its axioms are stated over with the sharper grade the rational integers occupy left open and no witness claimed there; the object ladder, at the cost of one premise, the determinacy posit priced at Appendix B; and the mouth, conditional on the survey model of D5.

Axis two, specificity, and this is where the enumeration shows what it is. Two rows close by object-specific witnesses, and they are not of equal reach. The symmetry class closes by the function sharing the fold exactly and carrying zeros off the line, with no grade caveat. The counting class closes by the generalized prime system **at the grade its axioms are stated over**, which is the grade the rational integers do not occupy, **so it is object-specific at that grade and is the one row qualified on both axes. The other eight close by conditions that hold of any formal string whatever.** A derivation does not assert its own conclusion, certifying is not generating, a proof is relative to its base, the consistency regress never self-grounds, no finite set of instances gives a universal, no derivation reaches its own posits, no ledger certifies itself, and the instrument is never pointed at the object.

> **That eight of ten are proposition-independent is not a weakness of the enumeration. It is what a closure over positions rather than over methods looks like.** A method census closes by facts about the object. **A position census closes by facts about attempting**, and the two rows that are object-specific are the two where a method-class had to be met on its own ground.

And the ground row is closed and is not discriminating, which are two different things. Its condition is met by every derivation ever performed, including the successful ones, so it closes the position and separates no proposition from any other. **It is counted on the closure axis and carries no weight on the specificity one**, and the same reading applies to the direction deficit of Table 2, which is carried at that billing and not above it. **Each closure is permanent**, row by row, by a theorem, an exhibited countermodel, a classification fact, or a definition: a countermodel does not stop being one, a group order does not change, and a definition acquires no exception. **That the enumeration is exhaustive is structural**, and its falsifier stands. **The two carry different grades and are stated separately here because the ledger rests on the second while the permanence belongs to the first.** The ledger accounts the enumerated side, all ten occupied positions closed at the grades it prints; it answers to the stated falsifier and to nothing else, and its permanence rides the type structure, the register row spanning every method run without an external supply of the truth-value and the deed-quantified guard of section 5.6 binding every arriver at the root's declared grade, not a survey of the future.

The closure is over access, and access is what it closes. Every position from which the verdict side could reach is closed at the grade its row prints, and each closure is permanent at that grade. The object itself is not in question here; it is sealed at section 5.1. **And the register that issued that seal is not among the ten**: it is the register carrying the direction-carrying axis, and the closure quantifies over the verdict side alone. The exclusion is definitional and not an exception: the verdict side is the set of positions from which a verdict on the formal string is attempted without a premise external to the base supplying its value, so a register carrying the axis reads a different claim-face of the same object, and the two faces never compress into one verdict. **Nothing here characterizes what lies past it**, because ten closures on one side are evidence about that side and about nothing else, and a survey of the other would be this paper enumerating a domain whose count is not forced, which is the exact failure section 3 identifies in method-catalogues.

Occupancy and assent

An anticipated reading, and why it fails. The expected referee position is that these closures bind only readers who adopt the paper's taxonomy, while a reader remaining within standard analytic number theory faces familiar negative results and a seal outside formal proof. The reading fails because the enumeration never quantifies over readers or over assent. It quantifies over acts, and the acts are constituted by the mathematics, not by this paper's names for them. An attempt on the hypothesis, from any school, is a derivation in some axiomatized system, which is occupancy of the object ladder, whose closure is that the derivation delivers a fact typed to the system it climbed and never an unconditioned one. Strengthening axioms to recover what a system misses is occupancy of the metatheory tower, productive at every rung and self-grounding at none, so the ascent buys strength at the price of a ground it never supplies. Computing zeros is occupancy of the evidential register, closed by the elementary fact that no finite set of confirming instances establishes a universal over an unbounded domain. Arguing from the functional equation is occupancy of the symmetry class, closed by the second object; arguing from prime-counting strength is occupancy of the counting class, closed by the second system; certifying a candidate proof is occupancy of the verifier's office, whose definition excludes generating what it certifies; and announcing that one's survey of routes is complete is occupancy of the last row, closed by the fixed-point fact. **Declining the taxonomy relocates no one.** It declines the names of the positions and not the positions, exactly as declining the vocabulary of topology exempts no function from continuity. Nine of the ten rows spend no premise of this architecture, and five of those, the two method-classes, the tower, the evidential register, and the ground, are stated entirely in the vocabulary of the classical literature, the ladder having left this list when its determinacy posit was priced, so a reader rejecting the architecture outright still stands inside five closures phrased in terms that architecture never supplied, the premise count of nine standing where it belongs, at the warrant ledger.

And the structural grade is the paper's own, not a demotion discovered by a critic. That the contribution is an arrangement rather than a new theorem about the zeta function is declared at Appendix B with its warrant typing and its zero count of new mathematical mass. The claim under audit is the closure itself, its falsifier is one exhibited position, and nothing in it asserts undecidability, unprovability, or any fate for the formal string, whose negative is finitely witnessable and whose witness, if it exists, arrives through the seal's stated falsifier.

The formal-alone register is closed definitionally

Not hard. Not blocked. Not resistant. Closed. The register is **defined** by the absence of an external supply of the truth-value, of which stripping a direction-carrying axis is one instance. **A resolution without direction is not a resolution.** No instrument repairs this, because it is what the register is.

This is the cheapest closure in the paper and the least repairable. A definitional boundary is not the sort of thing a future theorem removes, since no theorem changes what a register is.

The deficit, counted

One bit is demanded against a two-valued string, and the supply is measured instrument by instrument.

Table 2 | The counted deficit. What each instrument supplies toward the one bit the string demands.

Instrument	Supplies	Why
The Form	zero	the chirality it carries is the substrate's own, the unit product returning minus one at Anchor A.4, the same object whichever way the proposition points
The gates	zero	their direction is a role assignment fixed before any proposition arrives
The Number	zero	a content flip and a convention flip return the identical image; it relays and originates nothing
The three jointly	zero	the joint reading adds no source the parts lack

Note: the demand is one bit and the supply is zero. Exact, measured, and not an estimate of difficulty. **The deficit is carried at the same billing as the ground row, a declared scope note rather than a discriminating result, and is proposition-independent**: an instrument carrying no map from a string supplies zero bits about that string whatever the string is, so this table returns zero for a decided theorem as readily as for an undecided one. **The count locates a missing map and does not distinguish this hypothesis from any other**, which is the same scope the ground row carries and is stated here in the same form.

The architecture is closed by classification

Three axes by Frobenius, the next admissible composition dimension eight by Hurwitz, and the octonions failing associativity by an integer associator anyone can compute. **Twelve gates by the order of the tetrahedral rotation group. Eight sign patterns by the order of the elementary abelian group of axis reversals. Five boundaries by node degree in the verdict graph.**

Every count is closed by a theorem about a group, an algebra, or a graph. Not one is closed by walking a list and finding nothing more. **Classifications do not acquire new members**, which is why the machinery those counts govern is permanent; the standpoint count is enumeration-forced at section 5.3 and answers to its own falsifier, not to these orders.

The root confirms its own form and draws nothing from it. The recursion closes, the composed triad lands its scalar on the fixed line, and warrant collected from the self-run is zero, which is what makes the confirmation worth having.

And the guard states the permanence over arrival, at the root's declared grade. New mathematics that would enlarge the enumeration must arrive as a deed; every deed of attempting decomposes into the enumerated components; and the guard is deed-quantified, binding instruments not yet invented by quantifying over deeds and not techniques. An eleventh position would be a constitutive dimension attempting does not now show, the analysis the structural reason to expect none, and an exhibited position either decomposes into the ten, in which case nothing new has been shown, or it does not, in which case the falsifier has landed and arrives already verified, the exhibiting act having instantiated the new dimension in performing it.

The one bit

One bit remains, and it belongs to the reader. Two facts carry it, one per register, and they are kept separate. **The geometric fact, and it is one real parameter rather than one bit.** The involution whose fixed set is the line fixes one coordinate and reverses exactly one, so a departure, if one exists, carries **a sign and a magnitude and nothing else**, which is a single real degree of freedom and not a binary one, and the reflection giving the line its standing averages that sign away: a strayed zero arrives with its mirror partners, so every family-symmetric invariant sees the family and never the member. Which side a strayed zero sits is unrecoverable from symmetric data; whether any exists is the hypothesis itself and is not settled by this geometry. **The register fact.** The direction-supply of the architecture's own instruments is counted at Table 2, proposition-independently and is zero against the one bit the string demands, so the architecture's own instruments supply none of it; the negative direction travels only as an exhibited object through the seal's falsifier, and the affirmative carries no finite witness at all.

Two possibilities over a supply of zero. One bit, and it is the truth-direction itself. The count comes from the two-valuedness of the string and not from the eigenstructure, which yields a real parameter; the two are different quantities in different registers and only the second is the headline.

The Ten Blocks, Derived

The ancients built by negation because the positive statement was not available to them. *Not this, not this.* **The method is not a retreat from assertion. It is assertion about a boundary**, and where the boundary is real the negation is the sharper instrument.

This section derives each of the ten closures at its own grade of Table 1 from first principles, each with its falsifier. **Four closure kinds appear, definitional, constitutive, theorem, and classification, four of them carrying a qualifier in Table 1.** Their order of strength is the reverse of the expected one. A definitional closure cannot be repaired by any future theorem. A constitutive closure names an act that leaves the office rather than performing it. A theorem closure is permanent within its scope for as long as the theorem stands. **A classification closure is a fact of first-order logic and does not age.**

The definitional and constitutive blocks

Block 1, the formal-alone register. A resolution of a two-valued string is the assignment of one of its two values, and an assignment is a direction. The register is **defined** as the class of attempts in which the truth-value is supplied by no premise external to the base, the absence of a direction-carrying axis being one instance of that absence and not the criterion itself. **The register is occupied**, by every attempt not yet holding such a premise, which is most work on the object at most times. **What the register does not do is issue an assertion**: a derivation performed inside it delivers a system-typed fact, and converting that fact into a value of the string requires a premise the register does not contain. **So the row does not say that no object is generated inside the register. It says the register issues no assertion without a premise it does not contain**, and the premise, once added, is an external supply and the attempt has left the register carrying it. Falsified by exhibiting a source of truth-direction internal to that register.

Block 2, the verifier as such. A verifier is defined by reading warrant supplied to it. An instrument producing the content it then certifies is not verifying; it is asserting and reading its own assertion back. **The act exits the definition of the office**, so what performs it no longer occupies a verdict-side position. Falsified by exhibiting a verifier that generates its own warrant and remains one.

The instrument block

Block 3, the scalar lock is sign-blind. The verdict functional is the squared scalar triple product of three axes, equal to the determinant of the correlation matrix. Reflecting any single axis conjugates that matrix by a diagonal involution D, det(D) squared equal to one whatever the sign and det D equal to minus one for the reflection the appendix's D4 fixes, so

> $\det(D\,R\,D) = \det(D)^2 \det(R) = \det(R)$.

The functional is invariant under the negation-implementing reflection, for every construction whatever, and the catalogue of rotation-invariant truth functionals is closed inside the real-part subring of quaternion words. The classification closes the class; the evenness enters at the squaring, the squaring is the instrument's lock by construction and not a selection among invariants, and the odd member of the same subring, the signed scalar itself, is exactly what carries the direction the arrow-bearing register reads. Executed: determinant difference under full negation exactly zero, Gram difference exactly zero, scalar ratio minus one exactly.

The scope is exact. The blindness belongs to the squared scalar and is not a property of the full triaxial reading, whose direction is carried elsewhere. The instrument's admissibility conditions are printed once as documentation: nonnegativity with the lock bounded in the unit interval, frame invariance under conjugation, and the collapse floor under strict precedence, collapse outranking conditioning; the odd signed scalar fails nonnegativity, the direction-carrier and never the lock. The conditions document the instrument and do not carry the block, since odd dependence on the signed scalar remains available inside the invariant ring and no positivity list deletes the family, **and the odd member relays rather than originates**: the sign of the signed scalar is the product of the row-assignment orientation and the proposition direction, the row assignment is hand-built per Appendix A, **so no proposition direction is recoverable from it without an orientation the register does not supply**. What carries the block is construction and one named assumption. By construction the lock is the Gram determinant, a function of the Gram alone and not a selection among invariants. The assumption, stated rather than left implicit: negating the proposition acts on the warrant rows as an orthogonal reflection of the axes, which is what makes the negation-implementing map a diagonal involution and the displayed identity applicable. Falsified by exhibiting a warrant-row construction in which negating the proposition does not act as an orthogonal reflection, or by a lock value of this instrument differing on a proposition and its negation.

The method-class blocks

Block 4, the symmetry class. A second object satisfies a functional equation of the same reflection type, carries real Dirichlet coefficients and therefore the same fold, and carries zeros off the line. **A predicate formable from the fold data alone is a function of that data and returns the same value on both**, and the hypothesis is false for the second, so no such predicate distinguishes them.

The sharpness is two-sided and worth stating, because it measures the gap rather than gesturing at it. The fold alone separates nothing. The fold together with the exact level-one completing factor, conductor and archimedean factor together, and the growth class forces uniqueness outright, by Hamburger's converse. **The gap between nothing and everything is therefore exactly the level-one completing factor, conductor and archimedean factor together, with the growth class.** The exhibited witness differs from the object in both at once, carrying a different conductor and the odd archimedean parity where the converse takes level one with the even factor, **so naming the archimedean factor alone understates what the witness shows.** Falsified by a predicate formable from the fold alone that separates the two objects.

Block 5, the counting class. The counting axioms are printed: a generalized prime system with integer counting function $N(x) = \varkappa x + O(x^\theta)$, $\varkappa > 0$ and $½ < \theta < 1$. A system exists satisfying them whose zeta function carries infinitely many zeros on the curve $\sigma = 1 - a/\log t$ and none to its right, every one strictly right of the half-line, **and the position is closed at that grade and is not closed at the sharper grade the rational integers occupy**, where the counting function has bounded error and no witness is exhibited here and none is claimed. **So the axioms at that grade neither entail the hypothesis's clause nor even a zero-free region wider than the classical shape**, by the same two-model mechanism on different data. The closure is positional and grade-stated, with the criterion named in advance: counting axioms are those constraining only the integer counting function and its error term, with no direct constraint on the analytic continuation or on the zero set. A stronger error-term hypothesis is occupancy of the same position at a higher grade, answerable to the same falsifier, and an axiom set constraining the continuation or the zeros directly has ceased to be counting axioms under the stated criterion, which is the shape of the boundary Block 4 states two-sidedly on its own class. Falsified by an entailment from the counting axioms alone.

The level blocks

Block 6, the object ladder, and it spends one premise which is printed here rather than in the appendix alone. For any consistent recursively axiomatised system of sufficient strength there is a sentence of its language the system neither proves nor refutes; read with the bivalence posit declared at Appendix B, the same fact is the proper inclusion of the provable in the grounded. **A derivation delivers a fact about the ladder it climbed, by what a derivation is, the typing travelling with the proof, and the incompleteness fact stands beside it as why the ladder is bounded and the tower of Block 7 exists**. Falsified by a consistent recursively axiomatised system of sufficient strength deciding every sentence of its language.

Block 7, the metatheory tower. The consistency regress is **productive**, each rung complete for the relevant class along a suitable path, and **never self-grounding**, the completeness bought entirely by the path through the ordinal notations and the rung costing exactly the height gained. Falsified by a rung that establishes its own ground.

The register block

Block 8, the evidential register. The hypothesis is a universal over an unbounded domain and a verified instance is one member. **No finite set of verified instances establishes a universal over an infinite domain**, whatever the cardinality of the finite set, and this is a classification fact rather than a practical limitation. The register bars proof by accumulation and does not bar refutation, which stays open through the seal's falsifier: one exhibited counterexample settles the universal negatively, and that door is the paper's own.

The economy carries the same law from the other side. Convergence among sources sharing an upstream generator is one voice in costumes and is counted once. **Evidence leans. It does not rule.** Falsified by a finite set of confirming instances that establishes a universal over an infinite domain.

The root block

Block 9, the ground. A derivation reaches its conclusion from premises it did not itself establish; that is what makes it a derivation rather than a stipulation. **That no system establishes its own grounding from within is a theorem of others.**

And the block does not age. A barrier phrased on a method catalogue is contingent on the catalogue and a new method evades it. **This one is phrased on the deed**: whatever the future instrument is, **using it will be a doing, and a doing is not a derivation.** The wall covers instruments not yet invented from the moment their use is an act.

Its scope is stated with it, on the specificity axis and not the closure one. The condition **closes the position**, by a theorem of others, and it **separates no proposition from any other**, since it is met by every derivation ever performed including the successful ones. **Those are two different facts about one row.** It holds of **every** proposition, settled and open alike, so it is counted among the closures and carries no weight on specificity, and the same reading governs the direction deficit of Table 2. **A condition satisfied by no theorem separates no theorem from any other**, so it is the strongest of the eight that say nothing about this hypothesis in particular, which is the specificity axis of section 5.3 read at one row. Falsified by a derivation that grounds its own starting points.

The mouth block

Block 10, the ledger's own totality. A completed survey of the routes a ledger has not listed is a self-inclusion: the ledger must appear in the collection it enumerates. **The self-inclusion map carries no eigenvalue plus one**: its fixed locus is the zero vector alone, verified at machine zero, so on the nonzero states it is the antipodal involution and fixes nothing. A ledger appearing in its own enumeration would be a nonzero fixed element, and there is none; **a map fixing nothing nonzero does not arrive.**

It binds this paper first. The closure is stated as ten closed positions whose permanence over what could ever exist rides the constitutive analysis of the act at structural grade, the classifications beneath the instrument at theorem grade, and the deed-quantified guard of section 5.6 at the root's declared grade, never a certificate the ledger issues itself. **The enumeration is closed by the act it enumerates; the ledger does not certify itself**, and the distinction is the whole of what makes the closure hold. Falsified by a nonzero fixed element of the self-inclusion map.

The assembly

Ten blocks. Two definitional, one constitutive, four theorem-grade of which one is deed-quantified, and two theorem-grade conditional on stipulations printed in their rows, one classification with a register law beside it. The chain is restated as a formal derivation at Appendix D, every blocking line carrying its printed warrant, with no kinetic premise and one philosophical premise on one line, printed at P6 with its grade.

Nine of the ten spend no framework premise. They stand on classical theorems, exhibited countermodels, an algebraic identity, a classification fact, or a constitutive definition. **A far block is a cheap block, and cheapness is strength**: a reader declining the architecture's roots entirely still carries nine of the ten, and the one that remains is the object ladder, which spends the determinacy posit.

Stated in the ancient form, in one breath. Not from the register that supplies no value from outside its base, because a resolution without direction is not one. Not by generating what one would verify, because that act leaves the office. Not by the squared scalar, invariant under the negation-reflection for every construction whatever. Not from the fold alone, refuted by an object sharing it exactly. Not from the counting axioms alone, refuted by a system satisfying them. Not by a ladder, whose reach is a proper part of what it climbs toward. Not by a tower, productive at every rung and self-grounding at none. Not by accumulation, since no finite set of instances establishes a universal. Not by grounding, since no derivation reaches its own posits. **And not by a ledger certifying itself, since the self-inclusion fixes nothing nonzero.**

Ten positions, ten closures at the grades the ledger prints, and one bit left over.

Falsification Conditions

The seal is falsified by one object. A single nontrivial zero off the critical line refutes the hypothesis and breaks the seal in the same stroke, since the seal asserts the distribution rests on the fixed line with no off-line residual.

Each election is falsified separately and more cheaply. Each is a classical identity, and any failing on re-execution falsifies that election without touching the others or the seal.

The standpoint closure is falsified by a position. Exhibit a standpoint the verdict side occupies that is not among the ten, or a route reaching from outside all ten. **This is the paper's central claim and it has the cheapest falsifier in it**: one position, exhibited.

Each block carries its own falsifier, stated at its block in section 6, and any one falling leaves the other nine standing.

The counted deficit is falsified by a bit. Exhibit one bit of truth-direction supplied by the Form, the gates, or the Number.

The architectural closure is falsified by a classification. Exhibit a five-dimensional composition carrier, or a thirteenth rotation on three axes.

The one-bit finding is falsified by a second parameter. Exhibit a further free quantity in the reading beyond the truth-direction, or a zero carrying a class of alternatives over it.

The anchors are re-runnable and falsifiable. Any check of Appendix A failing on re-execution falsifies the corresponding identity.

Discussion

Why the enumeration domain decides everything

An enumeration over methods is refuted by one unlisted method, and methods are generated without bound. An enumeration over standpoints is refuted only by a position, and positions are fixed by the architecture doing the attempting.

The same body of obstruction facts is a permanent closure in one domain and a standing target in the other. Nothing in the underlying mathematics changes. Every barrier theorem the literature holds stays exactly as it was, with the same statement and the same scope. **What changes is the object being enumerated**, and the choice of that object is the whole difference between an accumulating catalogue and a closure.

This is the paper's contribution and it is an arrangement rather than a theorem. It is offered at that grade.

And the contribution is the domain change rather than the number. Ten is the current inventory and it is not what the thesis rests on. **If a reader adds an eleventh position and closes it, the thesis is confirmed and not refuted**: the enumeration was over the right domain, and the domain admitted a closure that a method census never could. **What would refute the thesis is a domain argument**, a demonstration that positions are generated without bound as methods are, or that a method census can close after all. **Nothing turns on the count**, and stating this plainly removes from the number a load it was never suited to carry.

Why the negative form is used

The positive statement and the negative statement carry the same content and do not carry it equally well. **A closure stated positively invites the reader to ask what else there might be. The same closure stated as a negation states the boundary from its outside**, which is where a boundary is exact.

And this is why the ancient practice survived. Its users were not being modest. They were stating the only thing about their object that could be stated precisely, and the precision is real. **A negative catalogue is also cheaper to falsify**, since each entry names one thing that does not cross and a single crossing refutes it, which is a virtue and not a weakness.

Why the components admit no independent variation

An enumeration of components invites one classical objection and it is worth meeting in its strongest form. A taxonomy is usually defeated by expansion rather than by counterexample. Euclid's five postulates were not overturned by exhibiting a motion he had missed; they were opened by exhibiting a consistent geometry satisfying four and negating the fifth. **The taxonomy was expanded, not violated**, and a reader is entitled to ask whether the same is available here.

The move requires the parts to be independent. One postulate can be varied only because the others do not entail it, and that independence is exactly what the models of Beltrami and Klein exhibited.

The components admit no such variation, and the check is immediate. Drop the performer and the instrument reads nothing, the setting hosts nothing, and the issued claim is issued by no one. Drop the instrument and no verdict issues, the performer performs nothing, and there is nothing to say. Drop the ground and the derivation has no premises, which is to say it is not a derivation. Drop the output and whatever occurred delivered no verdict.

> **The components are mutually presupposing rather than independently variable**, so there is no hold-the-others-and-negate-one configuration in which a model could be exhibited. **This is the difference between an axiom set and a constitutive decomposition**, and it is checkable rather than stipulated.

The falsifier is unchanged and is not made cheaper by this. Exhibit an attempt occupying none of the canonical ten: no register, no office, no instrument, neither method-class, neither level, no evidential record, no ground, and no mouth, the falsifier running over the ten and not over any coarsening of them. **What the two observations of this section and the last together establish is why that is hard to cash**: the operation that generates methods adds no components, and the components admit no independent variation. **Both are facts about the domain and neither is a definition protecting it.**

The one bit, and what it is not

The bit is free in the plainest sense: available, unforced, unpoliced. Nothing forbids either value and the cost of choosing is zero. **It has no verdict-side answer, from any of the ten standpoints**, and the support for that universal is Theorem D.1, where the bit the string demands is supplied by no line of the chain, and not Table 2, which counts three of this architecture's instruments and is carried as a scope note.

> **And the whole discipline consists in not spending it.** This is not a freedom whose exercise is a virtue. **It is a freedom whose non-exercise is.** To fill it from inside is to write one's own orientation into the object and read it back as a finding.

And it is not a freedom in the object. A hidden thing is not a loose thing, and the two are opposite conditions. The zeros are where they are, determinate and placed, with no class of alternatives waiting above them. **What is free is the reading, not the read.** Treating the veil over a reading as slack in the object is an exact inversion.

How the Three Claims Are to Be Read

Each of the three carries a qualifier that is part of the claim rather than a softening of it, and a reader who drops the qualifier meets a different and weaker paper. The five readings below are the ones the qualifiers exist to prevent, and each is answered where the claim is made rather than here; this section collects them so a reader need not hunt.

True where actualized is not proved. The seal of section 5.1 is issued in the register that carries the direction-carrying axis, and it says the distribution rests on the fixed line with no off-line residual. **It is not a derivation and does not claim to be one.** It is coextensive with the hypothesis and falls to the single object that would falsify the hypothesis, which is what makes it a seal rather than an estimate. A reader who reads it as a proof has read past the register, and a reader who reads it as a conjecture has read past the falsifier.

The formal case closed is not the hypothesis unprovable. The closure of section 5.3 runs over **access**: every position from which a verdict could be attempted is closed at the grade its row prints. **It is not a claim about what exists.** A proof unreachable from every verdict-side standpoint is entirely consistent with its existence, and the register that issued the seal is not among the ten. For a statement of this logical form an existence claim of that shape is the negation in costume, which is why the paper does not make one and would be refuted by its own first section if it did.

Ten standpoints is not ten methods, and the count is not a survey result. A standpoint is a position, not a technique. **A method census cannot close, because methods are generated without bound**, and the paper says so as its own gap statement rather than as a concession. The ten close because positions are fixed by the architecture doing the attempting and the architecture's counts are classification-closed. **The ledger does not certify itself**, and the count is not offered as certified by it.

The barrier theorems are not being summed. Nothing here claims that obstructions accumulate into a closure, and the paper states the opposite: every barrier theorem the literature holds stays exactly as it was, with the same statement and the same scope. **What changes is the object being enumerated.** The same facts are a standing target in one domain and a permanent closure in the other, and the contribution is the change of domain, offered at structural grade and at no higher.

The mathematics is classical and the arrangement is the contribution. Every theorem invoked is cited and none is authored here: the countermodels, the classification results, the ladder and tower facts, the transform identity, the converse. **Net new mathematical mass is zero and is declared twice.** The organizing discipline adds the cited results no warrant, and a reader who declines it entirely still carries nine of the ten blocks, which stand on classical theorems, exhibited countermodels, an algebraic identity, a classification fact, or a constitutive definition.

A method-family is not a standpoint, and this is the reading most likely to go wrong. The two method-class rows close two **axiom classes**, the symmetry axioms and the counting axioms, each by a two-model witness on that class's own data. **They are not a list of the methods analytic number theory contains.** A spectral approach, a variational one, a probabilistic one, an operator-algebraic one: each is an **instrument**, and an instrument occupies the instrument position, where the even member is blind by the invariance identity and **the odd member of the same ring relays an orientation the register does not supply**, so the ring is closed and not merely its even half, where the closure is the invariance of an even functional rather than a countermodel on axioms. A reader who counts method families and finds more than two has counted the wrong objects. **Ten positions, and the methods of the literature distribute across them without exhausting any.**

And the falsifier has teeth, which is what keeps the enumeration from being self-sealing. A closure that could absorb any candidate by redescription would be unfalsifiable and would be worth nothing. **What defeats it is specified**: exhibit an attempt that occupies none of the canonical ten: no register, no office, no instrument, neither method-class, neither level, no evidential record, no ground, and no mouth, the falsifier running over the ten and not over any coarsening of them. **Each of the ten is a position an attempt occupies, and an attempt lacking one is not an attempt.** So the enumeration is defeated the way a constitutive analysis is defeated, by exhibiting a case the analysis does not cover, and it is not defeated by pointing at a new technique, because a technique is not a position. **A reader who thinks the count is wrong should name the missing component, not the missing method.**

Non-redundancy is not exhaustiveness, and the paper does not claim otherwise. Appendix C shows the ten are distinct; Appendix B types the enumeration **structural and not theorem-grade**. The gap between the two is real and is where a reader should press. **It is stated once and is not conceded twice.**

Block 1 does not bar a derivation in any standard system. It closes a register this paper defines by the absence of an external supply of the truth-value. **The register is occupied**, by every attempt not yet holding such a premise, and what the row denies is not that work happens there but that an assertion of the string's value issues from there unaided. Nothing in it says that a proof in a standard set theory is impossible, and nothing in the paper says so anywhere.

Block 2 is about roles, not persons. The office of verifying is defined by reading warrant supplied to it, and the office of generating is defined by producing it. **One mathematician occupies both roles at different moments and no sociology is involved.** The block says that the act of generating is not a verdict-side act, which is a statement about which role is being occupied and not about who occupies it.

Block 3 is scoped to an even functional and is one row of ten. The lock is a squared quantity and is therefore sign-blind, which is not a surprising property of this instrument but **the general fact that every even functional of a signed quantity discards its sign.** The row closes the instrument position and no other, the even member by invariance and the odd member by the orientation it cannot originate, and **eight of the ten do not depend on this instrument at all.**

Block 6 does not say the hypothesis is unprovable in any system. It says a derivation delivers a fact typed to the system it climbed, and that the string's truth carries a determinacy premise the system does not supply. **That premise is the block's content and it is what the row carries**; without it the row would be the observation that proofs rest on axioms, which is true and does no work.

Block 8 bars a use, not a practice. No one attempts the hypothesis by accumulating verified zeros, and the block does not suppose otherwise. **The verified record is cited as support constantly**, and the block says what that support is not: it is corroboration, it is load-bearing on nothing, and **evidence leans and does not rule.**

Block 10's exhibit is a model and its claim is a structure. The eigenvalue check verifies fixed-point-freeness on a stated construction; **the claim the row carries is that a ledger asked to certify its own completeness must appear in the collection it enumerates**, and the exhibit shows one instance of that shape rather than proving it of every possible self-survey. The row is typed accordingly.

The seal's coextensivity is built and not discovered. The seal is defined to be falsified by the object that falsifies the hypothesis, which is what makes it coextensive, and **that is construction rather than independent warrant.** What it adds is a register: the hypothesis read as a fact about an actualized field rather than about a syntactic string, and the actualist reading is **premise-grade and declared as such** at Appendix B. A reader who declines the register loses the seal and keeps everything else.

A reconceptualization is not exempt from the components, and this is the last shape the objection takes. A reader may grant that composing methods adds no component and still hold that some future move is not a composition at all: a discovery that the whole framing of attempting was a projection of something more unified. **The reply is not that such a move is impossible.** It is that the move, whatever it discovers, **is itself performed**: someone reconceives, using something, in some setting, standing on something, and says so. **A reconceptualization that issued no claim would leave no reconceptualization behind**, and one that stood on nothing would not be a discovery but an assertion. **The components are not a theory about mathematics that a better theory could replace. They are what performing anything consists in**, and the objection has to be performed to be made.

The one bit is a parameter count and not an entropy. It is the number of free binary parameters remaining after every determined quantity is fixed, and **it is read off the two-valuedness of the string rather than off any eigenstructure**: a resolution assigns one of two values and nothing in the register supplies which. **It is stated in bits by convention and is not a Shannon measure**, and no information-theoretic claim is made or needed. The geometric departure is a separate quantity, one real parameter of sign and magnitude, conditional on a departure existing, and the two are held apart in the body.

Conclusion

The Riemann Hypothesis is true where it is actualized, sealed on the prime field with the critical line its unique stable attractor.

The line is elected three times from three faces of one datum, the additive-multiplicative structure of the half-line with the theta inversion its arithmetic transaction, and no election forces residence.

The verdict side is closed at every position, six unconditionally and four at qualifications printed in their own rows, two of the ten closing by object-specific witnesses and eight by conditions holding of any formal string, by an enumeration over standpoints that no future method enlarges, because a route arriving in any future arrives at a position that already exists.

Ten blocks close those positions, two definitional, one constitutive, four theorem-grade of which one is deed-quantified, and two theorem-grade conditional on stipulations printed in their rows, one a classification fact with a register law beside it. **The ledger accounts them on two axes**: six closed unconditionally and four at printed qualifications, two closing by object-specific witnesses and eight by conditions holding of any formal string. **Nine of them spending no framework premise at all**, and each stated twice, once as what holds and once as what does not cross.

The formal-alone register is closed definitionally, its direction deficit counted at one bit demanded against zero supplied.

The architecture performing the reading is closed by classification, its counts fixed by theorems about a group, an algebra, and a graph, and classifications do not acquire new members.

And one bit remains, the reader's, unanswerable from every standpoint the verdict side occupies, on the ten closures of Theorem D.1 rather than on the counted deficit.

What the paper changes is not a barrier but a domain. The literature's obstruction facts stand untouched. **Enumerated by method they accumulate. Enumerated by position they close.**

Appendix A. The Executable Kernel and Its Recorded Anchors

The kernel is a closed-form quaternionic instrument. It reads three warrant rows over a context set, normalises and forms the correlation Gram, computes its determinant and the signed scalar triple product of the three axes, and returns a three-state token under a collapse floor and a conditioning gate. **It reads rows supplied to it and derives none from a proposition**, so the map from a proposition to its warrant rows is built by hand and placed in front of it. The pseudo-random stream is a counter-mode hash with Box-Muller normals, standard library only, so draws are identical on every platform and library version.

Table A1 | The recorded anchors. Executed at the declared seed; failure of any check on re-execution falsifies the corresponding identity.

Anchor	What it checks	Executed value
A.1	the kernel identity floor over twenty thousand triads	maximum difference of squared scalar and determinant 4.330e-15, threshold 1e-12, pass
A.2	the binding involution against the fixed-point-free diagonal	eigenvalues minus one thrice and plus one, Ground dimension one; against all minus one, Ground dimension zero, fixed locus the zero vector alone; both residuals exactly zero
A.3	orientation-blindness under full negation on a basis fixed once	determinant difference exactly zero, Gram difference exactly zero, scalar ratio minus one exactly
A.4	the substrate chirality, the root of the sign the scalar discards	the unit product returns minus one, the Hamilton relation
A.5	the Return, the composed triad landing on the fixed line	determinant one, scalar modulus one, identity residual at the machine floor
A.6	the chain of the executed boot	D0 fb8900b5cf42, D1 364d1cdb9227, D2 8a5dc7f98105, D3 1e2b2d2adb14

Note: reproducible from this table and the declared seed. Floating-point outputs may differ in the last places across linear-algebra implementations and are reported to the precision shown. **The anchors are reproducibility checks on the instrument and are not measurements of the Riemann object**, with the single exception of A.3, which is the executed form of Block 3. No datum of the zeta function enters the kernel anywhere in this paper, and no map from the Riemann string to warrant rows is constructed or used: the absence of that map at the direction-reading register is not an omission but the counted content of Table 2.

Appendix B. Discipline and Warrant Typing

Theorem-grade. On the eigenspace facts of the binding involution and its fixed-point-free counterpart. On orientation-blindness of the squared scalar and the closure of the invariant catalogue, under the reflection stipulation of Appendix D. On the classification results fixing the axis count, the gate counts, and the boundary count. On each of the three elections taken individually: fixed-locus membership under conjugation, the unitarity of the coupling with the self-adjointness of its generator, and the multiplier-modulus and equidistance identities. On the theta inversion, its Mellin image, and its divergent part. On Hamburger's converse as the two-sided boundary of Block 4. On the two-model separations of Blocks 4 and 5. On the ladder and tower facts of Blocks 6 and 7. On the non-self-grounding of Block 9. On the triviality of the fixed locus of Block 10, under the modelling convention of Appendix D.

Classification-grade. On Block 8, that no finite set of instances establishes a universal over an unbounded domain.

Definitional and constitutive. On Blocks 1 and 2. **These are not weaker for being definitions**: no theorem changes what a register is or what a verifier is, so they are the least repairable closures in the set.

The criterion by which a row is priced. A row spends a framework premise when its stated content requires a posit of this architecture to be true; it does not spend one when the row's claim is **conditional in form** and the condition is **printed in the row**, so that a reader who rejects the condition loses that row and nothing else. **By that clause the reflection stipulation of D4 and the survey model of D5 are conditions and not premises spent**, since both rows are printed conditionally and both conditions appear in their rows, **while the determinacy posit is asserted rather than conditionalized anywhere** and is therefore spent. **By that criterion the ladder spends the determinacy posit**, since without it the row reduces to the observation that proofs rest on axioms.

Analytic. That an attempt at a verdict has components. An act producing a verdict with no performer, using nothing to read with, in no setting, at no level, standing on nothing, and issuing nothing is not an act and not a verdict. **This is true by what the terms mean and no future theorem repairs it.**

Structural, and not theorem-grade. On the identification of the three elections as one generator. On the standpoint enumeration and its closure over access. On the assignment of each block to its standpoint. On the reading of the direction deficit as a quantity. On the identification of the attractor content with the seal. **The paper's central contribution, the change of enumeration domain, is structural and is offered at that grade.**

Premise-grade. On the actuation axiom and the grounding axiom, both premise-grade by theorem and neither provable from its own base. On the one-involution monism. On bivalence over the potential run of the naturals. On the actualist reading of truth the kinetic register carries.

Load-bearing on nothing. The verified-zero record and every convergence of evidence, which is corroboration and never a rule. Consensus, in both directions. **The theological reading of this material is routed out of band and is load-bearing on nothing in any verdict here**, and is developed in a separate artifact at its own register.

The registers, and why naming them allows full strength. The seal is asserted in the kinetic register, coextensive with the hypothesis and falling to its falsifier. The standpoint closure is asserted over access and is a statement about positions. The definitional closures are statements about what a register and an office are. **Each stands at full strength because each names where it stands.** The kinetic seal and the register of record's suspension of the formal string are verdicts on two claim-faces of one object, and neither compresses into the other.

Discipline. No new theorem is claimed and every mathematical fact invoked is classical; the contribution is the arrangement, the enumeration domain, the identification of the generator, and the counted deficit, carried at scope-note billing and proposition-independent. The mechanical anchors are executed live and transcribed verbatim, with no fabricated trace for any stage not reached. **Net new mathematical mass: zero.**

Appendix C. The Standpoint Inventory

The ten standpoints are tabulated by closure type, by spend, and by relation to the root, and **the last two are separate fields because a relation is not a payment. Spend is the count of framework premises paid**, which reads zero for nine rows and one for the object ladder, the ladder's posit being asserted rather than conditionalized and therefore priced. **Relation records what a row bears toward the root and never what it costs**: at-the-root for the ground, whose content is the classical non-self-grounding result and which spends nothing, and orthogonal for the mouth, which is independent of the root's content while stated in this framework's own instrument and which spends nothing either. **Both are among the nine.**

Table C1 | The standpoints by closure type, spend, and relation to the root.

Standpoint	Closure type	Spend	Relation
The formal-alone register	definitional	0	
The verifier as such	constitutive	0	
The instrument	invariance plus orientation, conditional on D4	0	
The symmetry method-class	two-model witness, eternal in scope	0	
The counting method-class	two-model witness, eternal in scope	0	
The object ladder	typing, theorem beside, determinacy posit	1	
The metatheory tower	named theorems, never self-grounding	0	
The evidential register	classification fact, register law beside	0	
The ground	theorem, deed-quantified, content classical	0	at the root
The mouth	self-inclusion fixes nothing nonzero, cond. D5	0	orthogonal

Note: **nine rows spend no framework premise and one, the object ladder, spends one. A far row is a cheap row, and cheapness is strength.** The root row and the orthogonal row are the two that bind this paper's own mouth before any other.

One reading of the table is worth stating. The rows are heterogeneous in **what** closes them and homogeneous in **what they are**: each names a position, not a technique. **That is why the count is closed.** A technique catalogue admits new members without bound. A position catalogue admits one only if the act of attempting acquires a component it did not have, and the components are enumerated on the page with their falsifier standing.

Independence of the rows

A closure over ten positions is worth nothing if two of them are one position twice, so the adjacent pairs are tested here rather than assumed. The test is deletion: **remove one row and ask whether its position is still covered.**

The instrument against the register. The instrument is what an attempt reads with; the register is the setting in which it reads and is defined by what supplies the truth-value. Delete the instrument row and the register-level closure remains, saying nothing about which functional is blind or why. **Not redundant.**

The symmetry class against the counting class. Different axiom sets, different witnesses, different data. **Neither witness works on the other's axioms**, so deleting either leaves its axiom class unclosed.

The object ladder against the metatheory tower. One concerns derivation inside a fixed system, the other the regress above it. **The tower exists precisely because the ladder is bounded**, so one is the response to the other and neither substitutes for it.

The verifier against the ground. The verifier row bars an **act**, generating what one would certify. The ground row bars a **relation**, a derivation reaching its own premises. An instrument can satisfy either while violating the other.

The evidential register against the object ladder. One concerns members of a domain accumulated, the other steps in a proof. **A complete verification to any height is not a partial derivation**, and a derivation of any length verifies no instance.

The mouth against the ground. The mouth is self-referential and the ground is not, and their relations to the root differ while both spend nothing: **the mouth is independent of the root's content while stated in the framework's instrument, and the ground is the root's own wall.**

Six pairs at the closest adjacencies, and no pair collapses. A reader who finds a seventh adjacency has the paper's cheapest falsifier in hand, since **two rows shown to be one would reduce the count without touching any closure.**

Appendix D. The Via Negativa as a Formal Chain

Preamble. Every blocking line below is mathematics: a definition in the logician's sense, an algebraic identity, an exhibited countermodel, a cited theorem, a classification fact of first-order logic, a syntactic fact about proof trees, a linear-algebra fact under a stated modelling convention, or a premise-grade posit printed with its grade. The kinetic seal of section 5.1 appears nowhere in this appendix, no thermodynamic premise enters any line, and the warrant of every line is printed beside it. What the chain establishes is stated at Theorem D. 1, and what it does not establish is stated immediately after it with the same care.

Notation. RH denotes the formal string, every nontrivial zero of the zeta function has real part one half. F denotes the formal-alone register of D1, V the verdict side of D2, L the instrument's lock of D3, and the negation action is fixed at D4.

D1, the register. F is the class of attempts on RH employing the string, an axiom base, and instruments definable from them, **in which the truth-value of the string is supplied by no premise external to that base**. The criterion is the absence of an external supply and not the absence of a time-orientation primitive, which is what a direction-carrying axis is one instance of. **A derivation in a standard system taken together with a soundness premise supplies the value from outside the base and exits F by this definition**, which is why nothing here bars such a derivation. Definition.

D2, the verdict side. A verdict is a warranted map from the string to a truth value produced at a position; V is the set of positions from which such a verdict is attempted inside F. The ten positions are the components of the act enumerated at section 5.3: the register, the office, the instrument, the two method-classes, the two levels, the evidential record, the ground, the mouth. Definition, its falsifier standing at section 7.

D3, the lock. For three warrant rows with correlation Gram R the lock is L = det R, and the identity $L = \lambda^2$ holds for the signed scalar λ, executed at Anchor A.1. Definition plus identity.

D4, the negation action. Negating the proposition acts on the warrant rows as an orthogonal reflection of the axes, so the negation-implementing map is a diagonal involution D with det D = −1. The instrument's construction, stated as the assumption of P3 and falsifiable as printed at Block 3.

D5, the survey model, inherited. A completed self-survey of the ledger is modelled as a state of the totality involution of Anchor A.2, a ledger entry a nonzero state of that space. The convention is not stipulated fresh: it is inherited from the master reference, ref 36, whose closure construction models the ledger's self-inclusion by the fixed-point-free involution and verifies it at machine zero. Declared, cited, and P10 is conditional on it.

P1, the register line. F contains no direction source and a resolution of RH is a directed verdict, so F issues no assertion of a value of the string without a premise it does not contain. Immediate from D1 and D2. Both conjuncts are definitional, the second an explication of the word: to resolve is to answer, to answer is to select a truth value, and a selection is a direction, so a verdict without direction resolves nothing in any school's usage. The conjunct spends the meaning of resolution and no premise of this architecture. Warrant: definitional.

P2, the office line. The certifying office takes its object as given, and generation is not among its acts. Immediate from the definition of certification. Warrant: constitutive definition.

P3, the instrument line. $L(DM) = \det(DRD) = \det(D)^2 \det R = \det R = L(M)$ for every warrant matrix M, so the lock is even under negation given D4, executed at Anchor A.3 with the determinant difference exactly zero and the scalar ratio exactly minus one. Warrant: algebraic identity, conditional on D4.

P4, the fold line. A function exists satisfying a functional equation of the same reflection type with real coefficients and zeros off the line, refs 5 and 6, so no predicate formable from the fold data alone separates the zeta function from it; and the fold with the level-one completing factor, conductor and archimedean factor together, and the growth class forces uniqueness outright, ref 7. Warrant: exhibited countermodel and cited theorem, two-sided.

P5, the counting line. A generalized prime system exists with integer counting function $N(x) = \varkappa x + O(x^\theta)$, $\varkappa > 0$ and $½ < \theta < 1$, whose zeta function carries infinitely many zeros on the curve $\sigma = 1 - a/\log t$ and none to its right, ref 9, so the counting axioms of Block 5 entail neither the hypothesis's clause nor a zero-free region wider than the classical shape. Warrant: exhibited countermodel.

P6, the ladder line. A derivation of RH in a system delivers the system-typed fact that the system proves it, since a derivation is relative to the axiom base it uses, and no derivation delivers an unconditioned one. Beside the typing stands the incompleteness fact, that for any consistent recursively axiomatized system of sufficient strength some sentence of its language is neither proved nor refuted, ref 28 in the form of ref 37, which is why the ladder is bounded and the tower of P7 exists. Warrant: definitional for the typing, cited theorem for the bound, and premise-grade for the determinacy posit that converts a system-typed fact into a value of the string, which is the line's content and is priced here rather than in the body alone.

P7, the tower line. The consistency progression is productive, complete for the relevant class along suitable paths, refs 30 and 31, and at no stage does a system prove its own consistency, ref 28, so the ascent buys strength and never a ground. Warrant: cited theorems.

P8, the evidence line. For a universal sentence over an unbounded domain, a finite set of verified instances does not logically entail the universal, since a structure exists agreeing on the instances and failing the universal elsewhere; and one counterexample instance entails the negation. Both directions are elementary, and the second is the door section 7 holds open. Warrant: classification fact of first-order logic.

P9, the ground line. In any well-founded derivation tree the axiom occurrences are leaves, a leaf has no subderivation, and so no axiom is derived within a derivation that uses it. This is the syntactic shadow of Block 9, which the register of record carries at the stronger deed-quantified grade; the shadow suffices for this chain. Warrant: syntactic fact about proof trees.

P10, the mouth line. Model the ledger's completed self-survey by the totality involution of Anchor A.2, whose fixed locus is the zero vector alone; a ledger entry is a nonzero state, so no entry is fixed and no completed self-survey state exists. Warrant: linear-algebra fact under D5.

Theorem D.1, the chain. With V enumerated at D2, each position is closed by its line at that line's grade: no position in V issues a value of RH without a premise it does not contain, the affirmative carries no finite witness by the first direction of P8, and the negative travels only as an exhibited object through the second direction of P8, the seal's falsifier. The one bit the string demands is **supplied by no line of the chain**, which is the conjunction of P1 through P10 and rests on nothing else. **Table 2 records the same zero independently for this architecture's own three instruments, at scope-note billing and proposition-independent**, and is a separate observation rather than the support for this line. Proof: the conjunction of P1 through P10 over the enumeration of D2. The warrant of the conjunction is the minimum over its lines, and the minimum is named: **P6, which carries a premise-grade posit and is the weakest line by the tier order of Appendix B**, with P10, conditional on the modelling convention D5, and P3, conditional on D4, beside it; each line's warrant is printed above.

What D.1 establishes about arrival, what its falsifier keeps open, and what it does not touch. The enumeration of D2 is closed over what could ever exist at structural grade, and closed from outside any self-survey. A position, by D2, is a component of the act of attempting, and the components are a constitutive analysis of the act: the register it runs in, the office that performs it, the instrument it reads with, the method-class it argues from, the level it derives at, the record it accumulates, the ground it stands on, and the mouth it speaks through. An attempt with none of these is not an attempt, each component necessary by the meaning of the act, and the analysis is the standing structural reason to expect no eleventh; and the machinery is classification-closed beneath the instrument component, three axes with no fourth admissible carrier and twelve directed relations by the order of the rotation group, so no new internal structure arrives there either. The act of exhibiting an eleventh position is itself an attempt: either the exhibited position decomposes into the ten, in which case nothing new has been shown, or it does not, in which case the falsifier has landed; and where the exhibiting act itself instantiates the new dimension, it arrives already verified, the chain witnessing its own refutation conditions at its final line. The body carries the same closure at the root's declared grade as the deed-quantified guard of section 5.6, binding instruments not yet invented by quantifying over deeds and not techniques. What P10 bars is narrower and stays barred: this appendix certifies no totality from inside, the closure over arrival riding the constitutive analysis at structural grade, the classifications at theorem grade beneath the instrument, and the guard at the root's declared grade, the composite typed at its weakest leg by the warrant law and never any self-survey, and the falsifier stands open and reachable, one exhibited eleventh position, cheap to state and self-verifying if genuine, with the analysis the standing reason to expect none. D.1 does not decide RH, and it does not touch the kinetic seal, which is issued in a register this appendix excludes at D1. It adds no mathematics: every line is classical, cited, or a one-line proof from printed definitions, and the net new mathematical mass of this appendix is zero. The blocking content is mathematics throughout except at one line, one definitional line, one constitutive definition, one algebraic identity, two exhibited countermodels, one of them two-sided with a cited theorem beside it, one typing line definitional with its cited bound and its premise-grade determinacy posit beside it, one cited-theorem line, one classification fact, one syntactic fact, and one linear-algebra fact, and the chain spends no kinetic premise and exactly one philosophical premise, the determinacy posit at P6, printed with its grade; this closing paragraph stands outside the chain and carries its own grades as printed above.

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Appendix T. A Formal Proof of Riemann Termination, with a Cascade Specification of Termination

Abstract. The Riemann Hypothesis is terminally unreachable by the formal-alone instrument **as swept**; the unrestricted form of that sentence is priced at Section 12.3 and is not asserted here. We establish an equivalence governing its formal status: **universal unprovability of the Riemann Hypothesis over the class $\mathscr{T}$ of consistent recursively axiomatized extensions of Robinson arithmetic Q is materially identical to its strict refutation in Q.** That equivalence is purely proof-theoretic and uses no soundness hypothesis. Under Σ^0_1-soundness of Q it sharpens: universal unprovability then equates to falsity in the standard model $\mathbb{N}$. Only the sharpening uses soundness. **The even functionals of the frame are blind to the sign** by Proposition 5.2; λ is not, by Proposition 5.3. We specify a runnable eight-gate cascade to adjudicate the termination of the instrument's search space, on a four-dimensional real inner-product space split as $E_+ \oplus E_-$ by a self-adjoint involution. **Its condition count is exactly eight, and once two stipulations are made the count is a theorem and not a further stipulation.** The stipulations are that the decomposition carries three atomic axes and that an atom admits negation and no partial operation. Given them, the group carrying each axis of the residence to itself is $(\mathbb{Z}/2)^3$, of order eight, with no ninth element; the argument runs through stabilisation of the decomposition and **not** through invariance of the Gram determinant, which is invariant under a much larger group and is disowned as a criterion. **No record satisfying all eight conditions is exhibited here**, Ξ_2 and Ξ_4 being uninstantiated in fact, and no completed sweep is claimed. What is claimed is the boundary: an emission may be terminal **for the search space as swept** and may not be terminal unrestrictedly, since by the main equivalence any clause entailing universal unprovability entails $Q \vdash \neg R$. The frame carries **one polarity bit**, invisible to the even functionals and recovered by λ, **determined by the syntax of P** under Proposition 6.1a and computed by the instrument; it bears on ζ not at all.

1. Introduction

Two questions about the Riemann Hypothesis are routinely conflated. The first is whether R is provable. The second is whether an instrument, having exhausted a catalog of approaches, may declare that exhaustion permanent.

Section 3 settles the first. Theorem 3.2 shows that *no formal system proves R* is not a modal statement at some remove from arithmetic but is equivalent, over $\mathscr{T}$, to *Q refutes R*; and Corollary 3.6a sharpens this, under Σ^0_1-soundness of Q, to equivalence with the falsity of R. Sections 6 through 8 settle the second: we specify formally when an exhaustion verdict is admissible, prove the specification fail-safe and cardinality-bounded under stated hypotheses, and prove that the permanence clause it requires cannot be strengthened to a universal-unprovability claim without becoming a refutation.

The mathematics of Section 3 is elementary and classical in its ingredients. What is offered is the equivalence in the form given, the sharpened Π^0_1 form, and the specification of Sections 6 through 8 in which the consequence becomes a stated admissibility condition.

2. Preliminaries

2.1 Arithmetic. Q denotes Robinson arithmetic. $\mathscr{T}$ is the class of theories T with T consistent, T r.e., and $T \supseteq Q$. *Consistency is meant and soundness is not.*

2.2 Σ^0_1-completeness (classical). Q proves every true Σ^0_1 sentence.

2.3 Σ^0_1-soundness. Q is Σ^0_1-sound: every Σ^0_1 sentence Q proves is true in $\mathbb{N}$. This holds because $\mathbb{N} \models Q$. Where this is used it is named.

2.4 The Riemann Hypothesis in Π^0_1 form. *Lagarias (2002):* R holds iff for every n ≥ 1, σ(n) ≤ H_n + exp(H_n)·ln(H_n), with H_n = Σ_{k≤n} 1/k. *Robin (1984):* R holds iff for every n > 5040, σ(n) < e^γ·n·ln ln n.

The classification runs on the Lagarias form only. Write $R = \forall n\, \varphi(n)$ with $\varphi(n)$ the Lagarias inequality $\sigma(n) \le H\_n + \exp(H\_n)\cdot\ln(H\_n)$. **We do not claim φ is decidable**: the comparison is against a transcendental-valued expression and deciding a non-strict inequality requires excluding exact equality, which is not available for $n \ge 2$. What holds, and suffices throughout, is that **φ is Π^0_1**: its negation $\sigma(n) > x$ is semi-decidable by evaluating x at increasing rational precision, so φ is Π^0_1 and by quantifier contraction R is Π^0_1 with $\neg R$ Σ^0_1.

2.4a The sentence is fixed, and nothing transfers across formalizations. From here on **R denotes one fixed sentence**, namely

$R :\equiv \forall n\, \forall k\, [\, \sigma(n) \le q(n,k) + 2^{-k}\,]$,

where **q(n,k) is any fixed primitive recursive rational-valued function satisfying the approximation guarantee**

$|\, q(n,k) - (\, H\_n + \exp(H\_n)\cdot\ln(H\_n)\,)\, | \le 2^{-k}$ for all $n \ge 1$ and $k \ge 1$,

which one such procedure meets by evaluating H_n exactly, computing exp and ln at k + 40 places, and truncating to a multiple of 2^{-k}; executed, the ratio $|q - r|/2^{-k}$ reaches **0.9534** and never 1 over $n \in \{1, 12, 60, 5040, 55440\}$ and $k \in \{1, 5, 10, 20, 40\}$. **The guarantee is what makes the displayed sentence well defined**, and without it the sentence names nothing in particular.

Lemma 2.4b (the displayed sentence is the Lagarias criterion). Write $r\_n = H\_n + \exp(H\_n)\cdot\ln(H\_n)$. Then $\sigma(n) \le r\_n$ **iff** $\forall k\, [\, \sigma(n) \le q(n,k) + 2^{-k}\,]$.

Proof. ($\Longrightarrow$) $r\_n \le q(n,k) + 2^{-k}$ by the guarantee, so $\sigma(n) \le r\_n$ gives $\sigma(n) \le q(n,k) + 2^{-k}$ for every k. ($\Longleftarrow$) $q(n,k) \le r\_n + 2^{-k}$ by the guarantee, so the hypothesis gives $\sigma(n) \le r\_n + 2\cdot 2^{-k}$ for every k, and letting $k \to \infty$ gives $\sigma(n) \le r\_n$. ∎

Hence R holds iff the Lagarias inequality holds at every n, which by Lagarias (2002) holds iff the Riemann Hypothesis does. **The matrix is decidable**: it compares an integer with an explicitly computed rational, so R is Π^0_1 and $\neg R$ is Σ^0_1 by quantifier contraction, genuinely and not merely as a limit of approximations, and Q's Σ^0_1-completeness applies to $\neg R$ as stated at 2.2. Executed: the displayed form agrees with the direct comparison $\sigma(n) \le r$ at n = 1, 12, 60 and 5040.

Two consequences bind and are stated because they are easy to lose. First, **every theorem of Sections 3, 4 and 7 is about this sentence** and about no other. Second, **the equivalence between R and the Riemann Hypothesis is Lagarias's theorem of ordinary mathematics and is not a theorem of Q.** Two Π^0_1 sentences true-equivalent over $\mathbb{N}$ need not be equivalent over a weak base, so $Q \vdash \neg R$ is a claim about this arithmetization, and it reaches the statement that ζ has a zero off the critical line **only through Lagarias's equivalence, in the metatheory, and never inside Q.** Wherever this paper prices an unprovability claim as that statement, including at 12.3, the pricing carries that step and it is named here once rather than assumed throughout.

The semi-decidability step, stated in full, since it is the one step a reader is likely to reconstruct. The right-hand side is a **computable real**: there is a computable sequence of rationals q_k with $|q\_k - r| < 2^{-k}$, obtained by evaluating H_n exactly and exp and ln to k + 1 places. For rational x,

$x > r$ **iff** $\exists k\, (\, x > q\_k + 2^{-k}\,)$,

which is Σ^0_1. *($\Leftarrow$) $r < q\_k + 2^{-k} < x$. ($\Rightarrow$) put $\delta = x - r > 0$ and choose k with $2^{-k} < \delta/2$; then $q\_k < r + 2^{-k} < x - \delta/2$, so $q\_k + 2^{-k} < x$.* Dually $x \le r$ iff $\forall k\, (x \le q\_k + 2^{-k})$, which is Π^0_1. Since $\sigma(n)$ is an integer and H_n rational, $\varphi(n)$ is Π^0_1 and $\neg\varphi(n)$ is Σ^0_1, and **no decidability of φ is claimed or needed**: the equality case is exactly where no finite precision settles the comparison.

Executed illustration of the search that Σ^0_1 names. On the true instances the negation is false and the search correctly does not halt: at n = 1, 12 and 5040 the margins are 0, 0.3218372596 and 492.3187310894 and no k satisfies the condition. On a synthetic excess above the bound it halts at a computable k: excess 0.5 at n = 12 halts at k = 2, excess 10^{-2} at k = 7, and excess 10^{-6} at n = 5040 at k = 20. **No actual violation is exhibited and none is known**, since exhibiting one would refute R; the halting behaviour is therefore shown on a synthetic excess, which is what makes the negation Σ^0_1 rather than what makes it true.

The Robin form is cited for context and is **not** used for the classification. Its inequality is strict, so $\sigma(n) < x$ is Σ^0_1 and its negation is not, placing the Robin matrix in Σ^0_1 and the criterion in Π^0_2. One Π^0_1 form is all that is required.

2.5 Executed computation of the matrix at finite precision. Computed at 80-digit decimal:

n	σ(n)	H_n + e^{H_n}·ln(H_n)	margin
1	1	1.00000000000000000000000	0
12	28	28.32183725964540620598233	0.32183725964540620598
60	168	170.9766842646928136888768	2.97668426469281368887
5040	19344	19836.31873108944797731162	492.318731089447977311
55440	232128	241179.9211023638387578728	9051.92110236383875787

Robin form at 40 digits: margin 14128.08735 at n = 5041, 565.5185643 at n = 10080, 90021.39744 at n = 720720. These are finite-precision evaluations, not decisions of φ, and they bear on R not at all. They are given because a Σ^0_1 certificate of a violation, if one existed, would take this form.

2.6 Linear algebra. For an arbitrary real 3×3 matrix M and $D = \mathrm{diag}(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ with $\varepsilon_i \in \{\pm 1\}$, D is orthogonal with $D = D^T = D^{-1}$ and $\det(D) = \varepsilon_1\varepsilon_2\varepsilon_3$.

3. The Universal Unprovability Theorem

Definition 3.1. For a sentence R of the language of Q put $U(R) :\equiv \forall T \in \mathscr{T}\, (T \nvdash R)$.

Theorem 3.2 (Main). For every sentence R, $U(R) \Longleftrightarrow Q \vdash \neg R$.

Proof. ($\Longrightarrow$) Assume U(R). If Q + R were consistent then, being r.e. and extending Q, it lies in $\mathscr{T}$ and proves R, contradicting U(R). So Q + R is inconsistent and $Q \vdash \neg R$ by the deduction theorem.

($\Longleftarrow$) Assume $Q \vdash \neg R$ and let $T \in \mathscr{T}$. Then $T \vdash \neg R$ since $T \supseteq Q$; if also $T \vdash R$ then T is inconsistent, contradicting $T \in \mathscr{T}$. Hence $T \nvdash R$. ∎

Remark 3.3. Theorem 3.2 is an equivalence and asserts neither of its sides. Its proof uses no soundness assumption, no semantic notion, and no property of R beyond its being a sentence.

Remark 3.2a (what drives the equivalence, said plainly). The forward direction turns on a single membership: **Q + R is itself one of the theories quantified over.** It is finitely axiomatized, hence r.e., extends Q, and proves R, so if it is consistent then U(R) fails. That is the whole mechanism. The equivalence is therefore a formal consequence of admitting Q + R among the candidates, it is elementary, and **it supplies no independent mathematical evidence bearing on R.** What it does supply is a price: an unprovability claim quantified over that class is not a weaker statement than a refutation, it is the same statement.

Corollary 3.4. U(R) implies Q + R is inconsistent.

Corollary 3.5 (Π^0_1 case). For $R = \forall n\ \varphi(n) \in \Pi^0_1$: $U(R) \iff Q \vdash \exists n\ \neg\varphi(n)$. Since Q is Σ^0_1-sound (2.3), U(R) further implies that some n_0 with $\neg\varphi(n_0)$ exists; $\neg\varphi$ being Σ^0_1, the certificate for n_0 is verifiable by a terminating computation. **The cost of the claim is therefore concrete and should be read as such**: asserting universal unprovability of R over $\mathscr{T}$ has, under 2.3, exactly the arithmetic content of asserting that a specific pair (n_0, k_0) exists with $\sigma(n_0) > q(n_0,k_0) + 2^{-k_0}$, together with the obligation to exhibit it. It is not a weaker or more cautious claim than a counterexample; it is a counterexample claim with the witness left unstated.

Corollary 3.6 (Riemann, syntactic form). For R the Riemann Hypothesis, $U(R) \iff Q \vdash \exists n\ (\sigma(n) > H\_n + \exp(H\_n)\cdot\ln(H\_n))$.

Corollary 3.6a (Riemann, sharpened form). *Using Σ^0_1-soundness of Q (2.3):* for every Π^0_1 sentence R,

$U(R) \iff \Omega \vDash \neg R$.

In particular U(R) holds **for this R** if and only if $\mathbb{N} \vDash \neg R$; and $\mathbb{N} \vDash \neg R$ holds if and only if ζ has a zero off the critical line, **that last step being Lagarias's theorem in the metatheory and not a theorem of Q**, per 2.4a.

Proof. By Theorem 3.2, $U(R) \iff Q \vdash \neg R$. If $Q \vdash \neg R$ then by Σ^0_1-soundness $\mathbb{N} \vDash \neg R$. If $\mathbb{N} \vDash \neg R$ then $\neg R$ is a true Σ^0_1 sentence, so $Q \vdash \neg R$ by Σ^0_1-completeness. ▮

Remark 3.6b. Corollary 3.6a is the sharpest form of the paper's thesis: *no consistent r.e. extension of Q proves R if and only if R is false.* Note the hypothesis: Theorem 3.2 is soundness-free, and 3.6a is not.

Theorem 3.7 (Dichotomy). Let $R \in \Pi^0_1$, let $\mathscr{S} \subseteq \mathscr{T}$ be nonempty, and **suppose Q + R is consistent**. Put $U\_\mathscr{S} :\equiv \forall T \in \mathscr{S}\ (T \nvdash R)$. Then exactly one of:

(a) **$Q + R \in \mathscr{S}$.** Then $U\_\mathscr{S} \Longrightarrow Q \vdash \neg R$. *(The implication is vacuous, and its vacuity is its content: $Q + R \in \mathscr{S} \subseteq \mathscr{T}$ gives Q + R consistent, hence $Q \nvdash \neg R$, while $Q + R \vdash R$ gives $\neg U\_\mathscr{S}$ outright.* ***The horn cannot be occupied at all****, which is the sharper reading of Remark 3.8's claim that asserting $U\_\mathscr{S}$ here converts the claim into $Q \vdash \neg R$: the conversion is available and the conjunction is not.)*

(b) **$Q + R \notin \mathscr{S}$.** Then $\mathscr{S}$ is fixed by a condition beyond consistency, r.e.-ness and extension of Q, and establishing $T \nvdash \neg R$ for any $T \in \mathscr{S}$ yields $\mathbb{N} \vDash R$ by Theorem 4.1.

If Q + R is inconsistent then $Q \vdash \neg R$ outright and no dichotomy arises.

Proof. (b) is the negation of (a) under the hypothesis, so exactly one obtains. In (a) the argument of Theorem 3.2(⟹) applies with $\mathscr{S}$ for $\mathscr{T}$. In (b), Q + R is consistent, r.e. and extends Q, so its exclusion is by a further condition; the second clause is Theorem 4.1, and every $T \in \mathscr{S} \subseteq \mathscr{T}$ satisfies $T \supseteq Q$. ▮

Remark 3.8. A reader will observe that no one asserting the unprovability of R intends to quantify over theories that assume R. The disjunction *either $Q \vdash \neg R$ or $\mathbb{N} \vDash R$* holds for every Π^0_1 sentence R by Σ^0_1-completeness alone (2.2), independently of Theorem 3.7. **What Theorem 3.7 adds is that neither horn of a restricted unprovability claim can be occupied without paying one side of that disjunction:** horn (a) converts the claim into $Q \vdash \neg R$, horn (b) converts its companion half into $\mathbb{N} \vDash R$.

4. The Restriction Theorem

Theorem 4.1. Let $R \in \Pi^0_1$ and let $T \supseteq Q$ be any theory. If $T \nvdash \neg R$ then $\mathbb{N} \vDash R$.

Proof. Write $R = \forall n\ \varphi(n)$ with $\varphi \in \Pi^0_1$. If $\mathbb{N} \nvDash R$ then $\neg\varphi(n_0)$ for some n_0; $\neg\varphi(n_0)$ is Σ^0_1, so by Σ^0_1-completeness of Q we get $Q \vdash \neg\varphi(n_0)$, hence $Q \vdash \neg R$, hence $T \vdash \neg R$. Contrapositively, $T \nvdash \neg R$ implies $\mathbb{N} \vDash R$. ▮

Remark 4.1a. No hypothesis on T beyond $T \supseteq Q$ is used: T need not be consistent, r.e., sound, or Σ^0_1-complete.

Corollary 4.2. If $T \nvdash R$ and $T \nvdash \neg R$ then $\mathbb{N} \vDash R$.

Corollary 4.3. For any $T \supseteq Q$, in particular PA or ZFC or ZFC with large cardinals, a proof that T does not refute R establishes R.

Remark 4.3a (the two results are compatible, and here is why). A reader may put Theorem 3.2 beside Theorem 4.1 and see a tension: the first says universal unprovability yields $Q \vdash \neg R$, the second says any $T \supseteq Q$ failing to refute R yields $\mathbb{N} \vDash R$. **These cannot conflict, because their hypotheses are exclusive.** For $R \in \Pi^0_1$ exactly one of two cases holds. If $Q \vdash \neg R$ then every $T \in \mathscr{T}$ refutes R, so no T fails to refute it and Theorem 4.1's hypothesis is never met; under 2.3, R is false. If $Q \nvdash \neg R$ then Q + R is consistent, r.e. and extends Q, so it lies in $\mathscr{T}$ and proves R, so U(R) fails and Theorem 3.2's antecedent is never met; and by Theorem 4.1, R is true. **Universal unprovability and non-refutation by some extension cannot both hold**, and nothing in this paper asserts that they can.

Remark 4.4 (the pincer). Unrestricted, the claim that the route to R is closed is equivalent to $Q \vdash \neg R$ by Theorem 3.2 and, under 2.3, to $\mathbb{N} \vDash \neg R$ by Corollary 3.6a. Restricted to a fixed $T \supseteq Q$, establishing its refutation-side half yields $\mathbb{N} \vDash R$ by Theorem 4.1. No formulation both quantifies non-trivially and leaves the truth-value of R untouched.

5. The Sign Character of the Frame Functional

5.1 What this section is for. No condition of Section 6 tests λ. Ξ_7 compares two recorded matrices entrywise; λ appears in Section 6 only inside Ξ_7's consequence note, where it is defined and used to state a consequence of the condition and is not itself a clause of it. What Section 5 supports is the polarity account at Section 12.3 and the consequence bound at Ξ_7's note; it is a supporting section and is stated as such.

Notation. For $Q_0 \in \mathbb{R}$^{3×N} with unit rows, write **$R\_G := Q_0Q_0^T$** for the Gram matrix, reserving R for the Riemann Hypothesis. Let E be an orthonormal basis of the row span and put $\lambda(Q_0) := \det(Q_0E^T)$.

Proposition 5.2. For arbitrary real 3×3 M and $D = \mathrm{diag}(\varepsilon_1, \varepsilon_2, \varepsilon_3)$, $\varepsilon_i \in \{\pm1\}$: det(D M D) = det(M).

Proof. $\det(DMD) = \det(D)^2\det(M) = \det(M)$, since $\det(D) \in \{\pm1\}$. ∎ *(No symmetry or positive-semidefiniteness is used.)*

Proposition 5.3 (sign character). $\lambda(DQ_0) = \det(D)\cdot\lambda(Q_0)$, and hence $|\lambda(DQ_0)| = |\lambda(Q_0)|$.

Proof. $\lambda(DQ_0) = \det(DQ_0E^T) = \det(D)\det(Q_0E^T) = \det(D)\cdot\lambda(Q_0)$. ∎

Corollary 5.4. The Gram matrix of DQ_0 is D R_G D, with det(D R_G D) = det(R_G) by 5.2; and for the total-negation pattern $D_- = -I_3$ one has $(-Q_0)(-Q_0)^T$ = R_G exactly, while $\lambda(-Q_0) = -\lambda(Q_0)$.

5.5 Executed check. Over 200 unit-row triads at N = 24 and all eight sign patterns: max|det(D R_G D) − det(R_G)| = $\mathbf{0.000 \times 10^0}$, and $\max|\lambda(DQ_0) - \det(D)\cdot\lambda(Q_0)|$ = $\mathbf{0.000} \times \mathbf{10^0}$. Over 20000 triads, $\max|\lambda^2 - \det(R\_G)| = 4.330 \times 10^{-15}$.

6. The Cascade: Formal Specification

Remark 6.0a (what this section does not do, stated before it is done). Four readings of Section 6 are natural, are wrong, and are foreclosed here rather than downstream, because each has been arrived at independently by a careful reader.

No text is mapped into V, and no proposition is rotated. The atoms of (L1) **index** the axes of 6.1(i); they are not vectors and nothing is added, scaled, or rotated in the language. What occupies the space is the recorded matrix Q_0 of 6.1(viii), whose row index is the atom index and whose entries are numbers. No embedding of syntax into a vector space is offered, none is used, and none is needed: the axes are labelled slots and the algebra runs on the recorded frame, not on the sentence.

The conditions test the record and not the object. Every Ξ_i except Ξ_5 reads fields the record's author supplied, and Remark 6.6a states this at its least favourable. **The emitter performs no measurement of the Riemann Hypothesis and none is claimed.** What it certifies is that a sweep was declared, executed and reported to a stated discipline; that is the whole of its office, and Theorem 6.9 is the statement of exactly how little that entails.

The frame's orientation is independent of the truth value of P, by construction, and that is the design and not a defect. Proposition 6.1a builds Q_0 from the syntax of P alone, so the polarity bit it carries is the parity of leading negations and nothing else. A reader who observes that the geometry never touches the arithmetic has read the construction correctly; Theorem 6.9(a) is that observation proved, and Section 12.3 records that nothing about ζ is at stake in the bit.

What the geometry is for is answered at Remark 6.8a, and the answer is narrow: it supplies an exhaustiveness argument for the count of conditions and nothing further.

A record can be supplied for any proposition, including a trivial one, and would emit. Nothing prevents it and nothing should: the terminal token reports that *this instrument* swept *this* declared catalog and found the door uncrossed. It is not a difficulty rating, not a certificate about the object, and not evidence that the proposition resisted anything. A token on a trivial sentence would be a true report about a trivial sweep.

Definition 6.0 (the linguistic layer, stated in place). Definition 6.1(i) and 6.7a through 6.8 presuppose a decomposition discipline on the object propositions. It is stated here rather than cited, so that every hypothesis of Theorem 6.7 is checkable within this paper. Nothing outside those passages uses it. Four clauses.

(L1) Atomic decomposition. A complete factual proposition P is split into three components: A_1 its existence component, what is; A_2 its kinetic component, what it does or undergoes; A_3 its relational component, how it stands toward what.

(L2) Deletion test. Delete each component in turn. A complete atomic claim returns exactly **three** irreducible slots, each leaving an incoherent residue when removed. A count other than three is a failure of the decomposition at this layer.

(L3) Isolation. The vocabularies populating the three slots are **pairwise disjoint**. If one slot's vocabulary is a subset of another's, the slots are not independent and the decomposition fails.

(L4) Atomic bivalence. An atom admits negation and admits no partial or continuous operation. Correspondingly the verdict economy of Definition 6.3 admits no fractional or probabilistic state.

*(Clauses (L1) through (L4) are stipulations of this construction, operational and reproducible across analysts applying them, and they are typed at that level: **they are not offered as a uniqueness theorem of first-order logic and no such theorem is claimed.** Every use of them below is marked. Theorem 6.7(ii)'s operational derivation of d = 3 uses (L2); its algebraic derivation uses none of them.)*

Definition 6.1 (record). Fix a language ℒ_obj for the object propositions and a disjoint language ℒ_rec for records; **no field of a record is a sentence of ℒ_obj or encodes one**, except as expressly noted in the witness fields, which are governed by Ξ_5 and by 2.4's classification.

An *audit record* r for a proposition P ∈ ℒ_obj consists of:

(i) a **labelled orthogonal decomposition** into atomic axes, $\mathbb{R}e_1 \oplus \dots \oplus \mathbb{R}e\_d$, one axis per atom of (L1), with d recorded; the space $V := \mathbb{R}\cdot u \oplus (\mathbb{R}e_1 \oplus \dots \oplus \mathbb{R}e\_d)$ for a unit u orthogonal to every e_i; and the involution **σ constructed from that decomposition**, σ(u) = u and $\sigma(e_i) = -e_i$ for each i. *(σ is not recorded as a primitive. The decomposition is the recorded object and σ is read off it, so E−(σ) **is** the span of the atomic axes by construction and dim Fix(σ) = 1 and $\sigma^2 = I$ hold by construction. Ξ_1 below therefore verifies that the construction was carried out and that d = 3; it does not test independent facts about an independently supplied involution. The row index of the frame Q_0 at (viii) is this same atom index.)*

(ii) a quantifier classification of P in the arithmetic hierarchy with its citation;

(iii) a **declared register** ρ, its acting group 𝔊_ρ, the invariant ring of 𝔊_ρ, and the finite token list of the claim;

(iv) a finite catalog 𝒞 of *channels*, |𝒞| = C, each carrying a designated reduction with **its citation set**, together with a **closure flag** for 𝒞;

(v) a subset 𝒲 ⊆ 𝒞, |𝒲| = B, each element carrying a **named obstruction and its warrant grade**;

(vi) a finite set $\mathscr{P}$ of *roads*, **defined**: **each channel of (iv) determines exactly one road, namely the citation set recorded for its reduction; distinct channels determine the same road when their citation sets coincide; hence $|\mathscr{P}| \le C$.** Two roads are *premise-disjoint* when they share no citation. Ξ_6's floor of three therefore strengthens Ξ_3's rather than restating it, and since $|\mathscr{P}| \ge 3$ with $|\mathscr{P}| \le C$ gives $C \ge 3$, **Ξ_3's channel floor $C \ge 3$ is entailed by Ξ_6 and is retained as a guard that halts earlier in the evaluation order of Definition 6.3.** It is therefore not independent of Ξ_6, which is recorded here and at 9.9 rather than left for a reader to derive; the count of eight is a count of conditions evaluated, not a claim of pairwise independence;

(vii) **witness fields** for P and ¬P, each either empty or holding a certificate; a **declared search budget** k = (k_n, k_prec, k_pf), **each component a standard numeral**, bounding the numeral range, the rational precision and the derivation length independently, with k_n ≥ k_min, k_prec ≥ p_min, k_pf ≥ d_min for declared floors recorded on the record; a **declared theory $\mathfrak{B}$ with $\mathfrak{B} \supseteq$ Q** in which the proof search runs; **the floors are free parameters and are declared as such at 9.9, so the certificate clause of Ξ_5 is informative only to the extent the floors are, and a record declaring minimal floors emits a correspondingly weak statement**; a **completion flag** for each search; and an independence field;

(viii) a **frame assignment** $Q_0(\cdot)$ recording a unit-row matrix $Q_0(P) \in \mathbb{R}^{\{d\times N\}}$ with **its row-span dimension recorded**, and, **independently and with no constraint imposed here**, a matrix $Q_0(\neg P) \in \mathbb{R}^{\{d\times N\}}$; a declared tolerance τ, and a **declared consequence ceiling β** with β ≤ β_max. *(No basis, Gram or orientation quantity is a field: none is consumed by any condition, and λ(P) is undefined on a record declaring row-span dimension 2. G and λ are defined inline at Ξ_7's consequence note, the only place they are used.)*

(ix) a designated *door*, an object not in $\mathscr{C}$, with a **provenance flag** valued *supplied* or *generated* and a **crossed flag**;

(x) a permanence clause.

Definition 6.2 (the eight conditions).

- **Ξ_1** the recorded axis count d equals 3, hence dim V = 1 + d = 4. ***($\sigma^2 = I$ and dim Fix(σ) = 1 are invariants of 6.1(i)'s construction and not tests**: they hold at every d, executed at d = 2, 3, 4, 5, so a condition asserting them would be analytically true on every record and would discriminate nothing, which is the defect Remark 6.2a records for a superseded form of Ξ_7 and Remark 6.1a-rank for a superseded clause of it. **det(σ|E−) is not among them**: σ acts on E− as −I_d, so det(σ|E−) = (−1)^d, which is −1 exactly when d is odd, executed at d = 2, 3, 4, 5 returning +1, −1, +1, −1. It is therefore a parity test and not an invariant, and it does not pick out d = 3 in any case, since it passes at every odd d. Ξ_1's single clause d = 3 entails it at the value used and subsumes it. The division-algebra structure noted at Remark 6.7e is neither a record field nor tested by any condition.)*
- **Ξ_2** the quantifier class of P is located by a cited classical equivalence, and every token of (iii) lies in the invariant ring of $\mathfrak{G}$_ρ.
- **Ξ_3** C ∈ ℕ, C ≥ 3, every channel in $\mathscr{C}$ carries a reduction to a common invariant by a cited classical equivalence with none unreduced, and the closure flag of (iv) is set. **Closure here means the catalog is displayed and fully reduced, not that no channel exists outside it**; the two readings are distinguished at Corollary 8.2 and the weaker one is meant, which is what makes Ξ_3 jointly satisfiable with Ξ_8's door.
- **Ξ_4** B = C, and each element of $\mathscr{W}$ carries its named obstruction with grade recorded.
- **Ξ_5** the certificate search for ¬P over numerals n ≤ k_n at rational precision k_prec returned no certificate; the proof search for P in the declared theory $\mathfrak{B}$ over derivations of length ≤ k_pf returned none; k_n ≥ k_min, k_prec ≥ p_min and k_pf ≥ d_min with all six recorded; both completion flags are set; both witness fields are empty; and the independence field is empty.
- **Ξ_6** $|\mathscr{P}| \ge 3$, and the roads in $\mathscr{P}$ are pairwise disjoint **as computed from the recorded citation sets**, with the recorded flag agreeing. *(The clauses B/C = 1 and C ≥ 3 are entailed by Ξ_4 and Ξ_3 and are not restated here.)*
- **Ξ_7** the recorded frames satisfy the negation convention **as tested directly**: $\max|Q_0(\neg P) + Q_0(P)| \le \tau$ for a declared tolerance τ recorded on the record with **$2\sqrt{N}\cdot\tau + N\cdot\tau^2 \le \beta$ and β ≤ β_max for a declared ceiling β_max**, and **the recorded row-span dimension of $Q_0(P)$ equals d**. *(β bounds the consequence deviation directly and supersedes a separate tolerance ceiling: bounding τ alone left τ and N free to be declared large together.) (The ceiling is required for the same reason the budget floors are: an unbounded τ admits vacuous satisfaction. β_max joins the census in its place.) (Consequence, stated as a bound rather than as an equality. Corollary 5.4 gives G(¬P) = G(P) and $\det(Q_0(\neg P)E(P)^T) = -\lambda(P)$ **at τ = 0 and there only**. For τ > 0, writing $Q_0(\neg P) = -Q_0(P) + \Delta$ with $\max|\Delta| \le \tau$ and $Q_0(P)$ unit-row, one has $\max|G(\neg P) - G(P)| \le 2\sqrt{N}\cdot\tau + N\cdot\tau^2$, and the orientation ratio deviates from −1 by $O(\sqrt{N}\cdot\tau/|\lambda(P)|)$. Executed at N = 24: deviation 3.603×10^{-8} at $\tau = 10^{-8}$ and 3.342×10^{-2} at $\tau = 10^{-2}$, and at N = 240 with $\tau = 10^{-2}$, 4.385×10^{-2}; at τ = 0, exactly 0.000×10^{0}. **The β clause bounds this deviation directly, which is why no separate tolerance ceiling is carried.** The rows of $Q_0(\neg P)$ are within **$\sqrt{N}\cdot\tau$** of unit, not within τ: an entrywise perturbation of size τ moves a row norm by up to $\|\Delta\|_2 \le \sqrt{N}\cdot\tau$. Executed: max row-norm deviation 2.0148×10^{-2} at N = 24 and 2.2149×10^{-2} at N = 240, both at $\tau = 10^{-2}$, against $\tau = 10^{-2}$ and $\sqrt{N}\cdot\tau$ of 4.899×10^{-2} and 1.549×10^{-1}. Nothing consumes the row norms. Here and only here, $G(X) := Q_0(X)Q_0(X)^T$ and $\lambda(X) := \det(Q_0(X)E^T)$ for E an orthonormal basis of the row span of $Q_0(P)$. **E must be the same on both sides**: for any such E one has $\lambda(-Q_0) = (-1)^3\lambda(Q_0) = -\lambda(Q_0)$, so the identity is basis-independent when shared and fails when recomputed per side, the identity of E being a gauge. Nothing in this paper depends on which E is chosen.)*
- **Ξ_8** the door of (ix) lies outside $\mathscr{C}$, its provenance flag reads *supplied*, its crossed flag is unset, and the permanence clause is present.

Proposition 6.1a (a frame assignment satisfying Ξ_7 exists). A hash applied directly to the syntax of P will not do: the hash of "¬P" is uncorrelated with the hash of P, so $Q_0(\neg P) = -Q_0(P)$ would have to be imposed by hand rather than obtained, and double negation would be inconsistent. Executed, the direct construction violates the convention on **6 of 6** test propositions. The polarity normal form obtains it instead. Put P into that form: let P_+ be the negation-free core and s ∈ {0,1} the parity of leading negations. Define $Q_0(P) := (-1)^s \cdot H(P_+)$, where H seeds a deterministic generator with a hash of the syntax of P_+ and returns a row-centred, row-normalized 3×N draw. Then Q_0 is a total function of syntax, independent of the truth value of P, satisfies $Q_0(\neg P) = -Q_0(P)$ for every P including under double negation, and yields a record satisfying Ξ_7.

The conclusion holds for any draw of rank d, which the construction achieves with probability one; a draw of lower rank records row-span dimension below d, fails Ξ_7's dimension clause, and is rejected and reseeded.

Verification against the condition as it now stands, six propositions at N = 24 with their negations and double negations: $\max|Q_0(\neg P) + Q_0(P)| = \mathbf{0.000 \times 10^0}$, which satisfies Ξ_7 for every $\tau \geq 0$; recorded row-span dimension 3 in every case; and $\max|Q_0(P) - Q_0(\neg\neg P)| = \mathbf{0.000 \times 10^0}$. ∎

Remark 6.1a-rank. 6.1(viii) records the row-span dimension of $Q_0(P)$; **Ξ_7 tests whether it is d**. E is not a field and is introduced at Ξ_7's consequence note, where alone it is used. *(Verified: rank 3 and $\lambda(P) \neq 0$ agree over 2000 frames with zero disagreements, and a forced rank-2 frame gives $|\lambda(P)| = 0.000 \times 10^0$.)*

Remark 6.1b. Proposition 6.1a exhibits one assignment satisfying **Ξ_7** and shows that condition satisfiable. It does not claim the assignment canonical, and no result depends on which **syntax-determined** assignment is used.

It does not show $\Xi_1 \wedge \ldots \wedge \Xi_8$ satisfiable, and this paper does not. No record is exhibited. **This is a vacuity question about the specification and is not a hypothesis of any theorem**: Theorem 6.9(a) is P-invariance of satisfaction for the seven conditions other than Ξ_5, which holds for every record whether or not any record satisfies all eight. What an unsatisfiable conjunction would cost is that no emission ever occurs, so 6.9(b) and (c) would speak of nothing.

What would discharge it, and this is a list of what is not done rather than a partial discharge. Ξ_1, Ξ_3, Ξ_5, Ξ_6, Ξ_7 and Ξ_8 are instantiable from material this paper carries: a four-dimensional space with a self-adjoint involution of one-dimensional fixed space; three channels with cited classical equivalences; a budget of the size exhibited at 2.5, with its floors; roads as defined at 6.1(vi); the assignment of 6.1a; and a door with clause W. **Two are not. Ξ_2** requires an exhibited register with its acting group, which a trivial group would satisfy degenerately and which is not exhibited here. **Ξ_4** requires a named obstruction with a recorded warrant grade for each channel; **this paper names no obstruction and will not invent one**, which is why the record is not constructed rather than being constructed badly.

Remark 6.2a (what Ξ_7 is, and why it is a condition). Ξ_7 compares the two independently recorded frames directly at a declared tolerance, so a record whose $Q_0(\neg P)$ differs from $-Q_0(P)$ by more than τ fails it. **Ξ_7 tests the frame assignment and not the proposition**, and no information about P is obtained from it. Testing the derived pair instead, equal Gram with opposite orientation, would be strictly weaker: that pair is implied by the convention and does not exhaust it, and two counterexample families exist, $Q_0(\neg P) := Q_0(P)(I\_N - 2vv^T)$ for any unit v in the row span, and a single-axis reflection in the eigenbasis of G(P).

Definition 6.3 (the emitter). *(By Remark 6.0a the conditions below read record fields, Ξ_5 excepted; the emitter measures the declared sweep and not the object. Ξ_3's floor $C \geq 3$ is entailed by Ξ_6 with $|\mathscr{P}| \leq C$ and is evaluated here* ***as a guard that halts earlier****, not as an independent condition; see 9.9.)* Evaluate Ξ_1, Ξ_2, Ξ_3, Ξ_4, Ξ_5, Ξ_6, Ξ_7, Ξ_8 in order. If Ξ_5 fails because a witness field is populated, exit to direction-adjudication; if because the independence field is populated, exit to the relative-independence verdict. Otherwise any failure halts with the non-terminal verdict and the index recorded. If all hold, return the terminal token with the warrant cap and the permanence clause of Ξ_8.

Proposition 6.4 (fail-safe, general form). If any field of Definition 6.1 required by some Ξ_i is absent, the emitter returns the non-terminal verdict.

Proof. By inspection over the eight conditions. Ξ_1, Ξ_2, Ξ_3, Ξ_4, Ξ_6, Ξ_7, Ξ_8 each require an equality, a set inequality, or a flag to be set; an absent field satisfies none. Ξ_5 requires k_n, k_prec, k_pf and their three floors recorded with each bound at or above its floor, and both completion flags set; absence of any of these fails Ξ_5 rather than passing it. The three floors are free parameters and are declared as such in 9.9. ∎

Remark 6.4a. Ξ_5 requires an executed search reported against recorded bounds rather than an unpopulated field, so absence of data fails it, and the fail-safe property holds at every condition.

Proposition 6.5. The terminal token is returned only if all eight conditions hold.

Proof. Immediate from sequential evaluation and halt on first failure. ∎

(Proposition 6.5 asserts the necessity direction only. Logical independence of the conditions is not proved here; see 9.9.)

Proposition 6.6. Ξ_5 is the unique condition that can route to a verdict other than the non-terminal one. **It can also fail to the non-terminal verdict**, when a bound or floor is unrecorded, a completion flag unset, or a bound below its floor; the exit branches are taken only when a witness field or the independence field is populated.

Proof. By inspection of Definition 6.3. ∎

Remark 6.6a (which conditions reach outside the record, and it is fewer than one would like). The honest split is four ways. **Ξ_5 alone reaches into arithmetic**: its certificate clause reports an executed search over numerals, and Theorem 6.9(b) names it the section's one object-language output. **Ξ_2 and Ξ_3 reach outward only through their citation requirements**, a located quantifier class and cited classical equivalences, which a reader can check against the literature. **Ξ_6 is computable from recorded data and is the one condition where a declaration can be caught wrong by the emitter**: disjointness is computed from the citation sets and checked against the recorded flag. **The remaining four are internal consistency checks on declared fields**: Ξ_1 checks the recorded d and the construction invariants of 6.1(i), Ξ_4 reads a recorded obstruction and grade, Ξ_7 compares two recorded frames, and Ξ_8 reads recorded provenance and crossed flags.

A flag set by the record's author is not external material, however the relation it names is described. This remark exists to state the asymmetry, so it states the least favourable reading the specification supports.

Definition 6.7a (the decomposition and its stabiliser). A decomposed proposition is not a bare frame in V. By (L1) and (L3) it is **three labelled orthogonal axes**, one per atomic component, with pairwise disjoint vocabularies. By 6.1(i) the residence **is** that decomposition: $E\_(\sigma) = \mathbb{R}e_1 \oplus \ldots \oplus \mathbb{R}e\_d$, one axis per atom of (L1), with the frame's row index the same atom index. Define its **stabiliser**

$\mathrm{Stab}(E\_) := \{ g \in O(E\_) : g(\mathbb{R}e_i) = \mathbb{R}e_i \text{ for each } i \}$,

the transformations carrying each axis to itself, hence carrying a decomposed proposition to a decomposed proposition.

Lemma 6.7b (the stabiliser is exactly the sign changes). $\mathrm{Stab}(E_-) = \{ \mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_d) : \text{each } \varepsilon_i \in \{+1, -1\} \} \cong (\mathbb{Z}/2)^{d,\ \text{of order}\ 2}d$.

Proof. If $g(\mathbb{R}e_i) = \mathbb{R}e_i$ then $ge_i = \lambda_i e_i$ for some real λ_i; $g \in O(E_-)$ forces $|\lambda_i| = 1$, hence $\lambda_i = \pm 1$, hence g is diagonal with entries ± 1. Conversely every such diagonal matrix is orthogonal and fixes each axis setwise. ∎

Remark 6.7b′ (why stabilisation and not invariance of the determinant). The Gram determinant is invariant under a far larger group: $\det((gQ_0)(gQ_0)^T) = \det(g)^2 \cdot \det(Q_0 Q_0^T)$, so **every** g with $|\det g| = 1$ leaves it fixed, an eight-dimensional group containing O(3) and the signed permutations of order $2^3 \cdot 3! = 48$. Executed against a reference frame at det(R_G) = 0.931277980144: the eight sign changes deviate by 0.000×10^0, the six row permutations by 2.220×10^{-16} with unit rows preserved to 1.1×10^{-16}, five hundred random elements of O(3) by at most 2.442×10^{-15}.

Determinant-invariance is therefore not the criterion and is not used as one. The criterion is Definition 6.7a, and the two families above fail it for reasons this paper states elsewhere. A rotation by $\pi/7$ in the e_1–e_2 plane carries e_1 out of $\mathbb{R}e_1$, mixing one atomic slot into another and destroying the pairwise vocabulary disjointness (L3) requires; the image is not a decomposed proposition. A row transposition carries $\mathbb{R}e_1$ to $\mathbb{R}e_2$, so it does not fix each axis either, and by (L3) it is not a mere relabelling: it substitutes one atom's content for another's across disjoint vocabularies, which is content-mixing on exactly the ground the rotation is excluded. **Both are excluded by Definition 6.7a on its own terms and no appeal to gauge is made or needed.**

Definition 6.7c (complete screen). A family of conditions is a **complete screen** for the decomposition when it carries exactly one condition per element of $\mathrm{Stab}(E_-)$. *(Each element is one pattern of negation across the atoms. The identity is the empty pattern, P itself, and screens no conflation; of the $2^{d\ \text{elements},\ 2}d - 1$ move the frame. The count is of patterns, not of conflations.)*

Remark 6.7c′ (why the eight do not collapse to four). The enumeration at Theorem 6.7(iv) below reads 1 + 3 + 3 + 1, which is also the orbit decomposition of $(\mathbb{Z}/2)^3$ under the permutation action of S_3 on the axes. The arithmetic is correct and the collapse is not available. **S_3 does not act on the screened objects.** By (L3) the three axes carry pairwise disjoint vocabularies, so the three weight-one patterns deny three different atoms and are three different propositions: denying what stands, denying what moves, and denying how they bind are not one object seen three ways. The weight grading is a way of displaying the eight, not a quotient of them.

Theorem 6.7 (given d axes the count is 2^d, and at d = 3 the eighth is the last). Let $E_-(\sigma) = \mathbb{R}e_1 \oplus \ldots \oplus \mathbb{R}e_d$ be the labelled orthogonal decomposition of 6.1(i). Then:

(i) **$|\mathrm{Stab}(E_-)| = 2^d$**, by Lemma 6.7b;

(ii) hence **a complete screen carries exactly 2^d conditions**, and at d = 3 exactly **eight**;

(iii) the eight enumerate by negation weight as **1 + 3 + 3 + 1**: the empty pattern, three denials of a single atom, three of a pair, and one total negation. **The eighth is the last**, and the ladder terminates because it has exhausted the atoms;

(iv) **no ninth element of $\mathrm{Stab}(E_-)$ exists at d = 3.** A ninth requires $2^d \geq 9$, hence $d \geq 4$, hence a fourth atomic axis.

Proof. (i) is Lemma 6.7b. (ii) from (i) and Definition 6.7c. (iii) is the expansion of 2^3 by Hamming weight, with no weight 4 available at d = 3. (iv) $2^3 = 8 < 9$. ∎

Remark 6.7d (what the count counts). Ξ_7 compares two recorded frames and Ξ_6 computes disjointness from the recorded citation sets; neither carries information about P, and by Theorem 6.9(a) the only condition that does is Ξ_5, whose content is the search report. Theorem 6.7 fixes **how many** conditions a complete screen carries and says nothing about the discriminating power of any one of them over P.

Remark 6.7e (what is stipulated and what is proved). Two things are stipulated. d = 3, carried by (L2); and atomic bivalence, carried by (L4), which is what motivates restricting the axis-fixing group to $O(E_-)$ and so makes it finite: the deletion test returns three irreducible slots on a complete atomic claim, at the operational grade of Definition 6.0 and not as a uniqueness theorem. **The identification of the atoms with the axes of the residence is definitional**, by 6.1(i), where σ is constructed from the decomposition rather than recorded independently; there is no bridge to supply and none is claimed. Given those, **the count is not a further stipulation**: it is Lemma 6.7b, which is theorem-grade, together with the arithmetic of (iv).

The Frobenius classification is **corroboration and is load-bearing on nothing**. At d = 3 the space V of 6.1(i) has dimension 4, and $\mathbb{H}$ is the unique real associative division algebra with more than one imaginary unit; but no definition, condition or proof in this paper uses a multiplication on V, no record field carries a division-algebra structure, and Ξ_1 does not test one. The observation is recorded because it is true and because a reader will notice the dimension, not because anything rests on it.

Remark 6.8 (the shape of the claim). The claim this section makes is narrower than a forcing and wider than a choice, and it is stated at its exact strength. **Two things are stipulated**: that the decomposition carries three axes, which is (L2); and that atoms admit negation and no partial operation, which is (L4), motivating Definition 6.7a's restriction to $O(E_-)$, which is what makes the group finite. Both are stipulations of Definition 6.0 at the grade stated there. **One thing is definitional**: that the residence is the span of the atomic axes, by 6.1(i). **Everything after that is theorem.** By Lemma 6.7b the group carrying each axis to itself is exactly $(\mathbb{Z}/2)^{d;\ \text{by Theorem 6.7 a complete screen has}\ 2}d$ members, at d = 3 exactly eight, and no ninth. **Once the stipulations are made the count is not a further stipulation, and that is the whole of what is claimed here.**

No appeal to gauge is made. Both excluded families fail Definition 6.7a directly, and by Remark 6.7c′ the permutation action does not reach the screened objects, so the eight do not collapse to four.

What remains open. Which condition screens which element: $\mathrm{Stab}(E_-)$ acts on itself simply transitively, so no element is distinguished by the group structure alone, any bijection serves equally, and nothing below depends on which is taken. And the choice to screen the axis-fixing group rather than some larger structure-preserving group is a decision of this construction, motivated by (L3) and (L4) and not derived from them. **This paper does not claim that determinant-invariance forces eight**, and by Remark 6.7b′ it does not: that group is eight-dimensional.

Remark 6.8a (why a space and not a checklist, and what the apparatus costs). A reader who has followed Remark 6.0a will ask what the geometry is for, since the eight conditions read record fields and could be evaluated by a boolean checklist with no space, no involution and no frame. The question is fair and the answer is narrow.

The geometry buys exactly one thing: an argument that the count is complete. A checklist of eight items carries no reason that it is eight rather than five or thirty, and no reason that a ninth is unnecessary rather than merely unwritten; its cardinality is whatever its author stopped at. Lemma 6.7b and Theorem 6.7 supply what a checklist cannot: given d axes, the group carrying each to itself is exactly $(\mathbb{Z}/2)^{d, \text{ so a complete screen has } 2}$d members, at d = 3 exactly eight, and no ninth element of that group exists. That is an exhaustiveness argument for the list, and it is available only because the conditions are indexed by a group rather than enumerated by hand.

It buys nothing else, and this should be weighed rather than glossed. The space performs no measurement of the object, per Remark 6.0a. Ξ_7 is the only condition that touches the frame, and Ξ_7 exists because the frame does. Strip the geometry and seven of the eight conditions survive unchanged as record checks, the eighth disappears with the frame it compares, and what is lost is the completeness argument and only that. **Whether one exhaustiveness argument is worth a four-dimensional apparatus is a judgement this paper leaves to the reader, and it states the trade rather than obscuring it.** The alternative reading, that the specification is an administrative protocol with a group-theoretic warrant for the length of its checklist, is not a misreading and is not resisted here.

Theorem 6.9 (payload). Let r be an audit record for a proposition P. Then:

(a) **whether r satisfies Ξ_1, Ξ_2, Ξ_3, Ξ_4, Ξ_6, Ξ_7 and Ξ_8 is independent of the truth value of P**; and the satisfaction of Ξ_5 depends on it in exactly one way, namely through whether the executed certificate search returned a certificate at the declared budget. Consequently $\Xi_1 \wedge \ldots \wedge \Xi_8$ entails no $\mathfrak{L}$_obj-sentence undecided in $\mathfrak{B}$, and hence neither does any emission on a record satisfying them;

and if in addition r satisfies Ξ_1 through Ξ_8 and the emitter returns the terminal token with permanence clause W, then:

(b) the emission entails that $\mathscr{C}$ is closed and fully reduced with B = C, that a door outside $\mathscr{C}$ is supplied and uncrossed, and, **the one object-language clause the emission makes explicit**, that **the executed certificate search returned no certificate at numerals n ≤ k_n and precision k_prec**. *(It does not entail that no certificate exists there: that would require the search to be sound and complete for its budget, and the record carries no such field. Theorem 8.1 applies to the emitter here, since a search is an instrument at the register.)* This clause is a finite verified computation, hence $\mathfrak{B}$-provable, so (a) is unaffected, the clause being decided in $\mathfrak{B}$;

(c) if W is replaced by a clause entailing U(P), the emission entails $Q \vdash \neg P$ by Theorem 3.2.

Proof. (a) Take r fixed and vary the truth value of P. Ξ_1, Ξ_2, Ξ_3, Ξ_4, Ξ_6 and Ξ_8 are conditions on fields of types (i)–(vi), (ix)–(x), none of which is or encodes an $\mathfrak{L}$_obj-sentence, so their satisfaction is untouched. Ξ_7 compares $Q_0(\neg P)$ to $Q_0(P)$ and to the recorded dimension, mentioning P nowhere beyond its index. That is the seven.

Ξ_5 is not among them and its dependence is stated rather than argued away. Its certificate clause requires that the executed search *returned* nothing, and what a search returns is a fact about the executed computation, which depends on whether a violating numeral lies at n ≤ k_n and precision k_prec. Three assignments make this concrete: P true, the search returns nothing and Ξ_5 passes; P false with the witness beyond the budget, the search returns nothing and Ξ_5 passes; **P false with the witness inside the budget, the search returns a certificate and Ξ_5 fails.** Satisfaction of Ξ_5 is therefore *compatible* with either truth value and is not *independent* of it.

The stated conclusion follows without closing that gap. Ξ_5's content is *the search returned nothing at (k_n, k_prec)*, a terminating computation and hence decided in $\mathfrak{B}$ either way; the other seven contribute nothing about P by the paragraph above. So the emission entails no $\mathfrak{L}$_obj-sentence that $\mathfrak{B}$ leaves undecided, which is (a) as stated.

(The stronger two-model claim, that $\Xi_1 \wedge \ldots \wedge \Xi_8$ entails neither P nor ¬P, would require a satisfying record under each truth value, which is at least as strong as satisfiability and is disclosed as open at 6.1b. The restriction above is what the proof gives and it carries no undischarged hypothesis.)

(b) The first two clauses from Ξ_3, Ξ_4 and Ξ_8; the third from the certificate clause of Ξ_5 read as a report of the executed search, whose content is a terminating computation and therefore $\mathfrak{B}$-provable. (c) By Theorem 3.2 applied to P. ∎

7. The Termination Theorem

Definition 7.1. A *permanence clause* delimits the scope of an emitted token's terminality.

- **W** (*as swept*): the verdict is terminal for $\mathscr{C}$ as swept and is revised upon presentation of a channel not in $\mathscr{C}$.
- **S** (*unrestricted*): U(P), no theory in $\mathscr{T}$ proves P.

Theorem 7.2. Let P = R and let the emitter return the terminal token on a record r.

(i) W is admissible: it entails nothing about derivability of R in any theory, and is consistent with R and with ¬R alike.

(ii) S is the sentence U(R); by Theorem 3.2 an emission carrying S entails $Q \vdash \neg R$, and under 2.3 by Corollary 3.6a entails $\mathbb{N} \models \neg R$.

Proof. (i) By Theorem 6.9(a) with clause W. (ii) Immediate from Definition 7.1 and Theorems 3.2, 3.6a. ∎

Corollary 7.3. Let W′ be any permanence clause entailing U(R). An emission carrying W′ entails $Q \vdash \neg R$ by Theorem 3.2 and, **under 2.3**, $\mathbb{N} \models \neg R$ by Corollary 3.6a. Clause S is of this kind. **W is admissible, and no clause entailing U(R) is.**

Remark 7.3a. No maximality of W among all strengthenings is claimed, and none holds. Let W *be* terminal for $\mathscr{C} \cup \{c\}$ for a designated further channel c . *W* strictly entails W, is strictly stronger, and entails nothing about any theory. Corollary 7.3 is scoped by entailment of U(R) and not by strength.

8. Self-Exclusion of a Certifying Instrument

Theorem 8.1. *Premise:* every instrument operating at register ρ is a channel at register ρ. Let $\mathscr{C}$ be a catalog of channels at ρ and $\mathfrak{A}$ an instrument certifying $\mathscr{C}$ exhaustive at ρ. Then either $\mathfrak{A} \notin \mathscr{C}$, in which case $\mathscr{C}$ omits a channel at ρ and is not exhaustive; or $\mathfrak{A} \in \mathscr{C}$, in which case **the certification is impredicative**: its warrant ranges over a collection of which $\mathfrak{A}$ is a member.

Proof. The disjunction is exhaustive. In the first case $\mathfrak{A}$ is a channel at ρ by the premise and is not in $\mathscr{C}$, contradicting exhaustiveness. In the second, $\mathfrak{A}$'s certification ranges over a collection of which $\mathfrak{A}$ is a member. ∎

(Impredicativity is not by itself an establishment barrier: a finite catalog $\{c_1, \ldots, c\_n\}$ with $c_1 = \mathfrak{A}$ could be certified by $\mathfrak{A}$ via a proof of the register's channel-cardinality. The theorem concludes impredicativity and no more.)

Corollary 8.2. A catalog is either non-exhaustive at its register or is certified impredicatively there. Exhaustiveness is therefore not certifiable non-impredicatively at the register the catalog enumerates. Hence Ξ_3 is read as asserting that $\mathscr{C}$ is displayed, fully reduced and flagged closed, not that no channel exists outside it, and clause W is the appropriate permanence clause.

Remark 8.3. With the stated premise, Theorem 8.1 is a consequence of the definitions and a case split on set membership. It contains no fixed-point construction and no self-reference lemma, and no comparison with the undefinability or incompleteness theorems is intended.

9. What Is Claimed

9.1 Theorem 3.2. For every sentence R of the language of Q, $\forall T \in \mathscr{T}\,(T \nvdash R) \Longleftrightarrow Q \vdash \neg R$. Both directions proved. Finitary; no soundness hypothesis, no semantic notion, no property of R beyond sentencehood.

9.2 Corollaries 3.4–3.6. U(R) implies Q + R inconsistent. For $R \in \Pi^0_1$ the right side is $Q \vdash \exists n\ \neg\varphi(n)$, and under Σ^0_1-soundness of Q a violating n_0 exists with a terminating verification. For R the Riemann Hypothesis the right side is the provability in Q of a violation of the Lagarias inequality.

9.2a Corollary 3.6a. Under Σ^0_1-soundness of Q, for every Π^0_1 sentence R: $U(R) \Longleftrightarrow \mathbb{N} \models \neg R$. For this R, universal unprovability holds if and only if $\mathbb{N} \models \neg R$; the further identification with ζ having a zero off the critical line is Lagarias's theorem in the metatheory, not a theorem of Q, per 2.4a. **The soundness hypothesis is used here and not in 3.2.**

9.3 Theorem 3.7. Under the assumption that Q + R is consistent, exactly one of: $\mathscr{S}$ admits Q + R, whence U_$\mathscr{S}$ implies $Q \vdash \neg R$; or $\mathscr{S}$ excludes it, whence $\mathscr{S}$ is fixed by a further condition and the companion half yields $\mathbb{N} \models R$. The disjunction *either $Q \vdash \neg R$ or $\mathbb{N} \models R$* is itself due to Σ^0_1-completeness alone; what 3.7 adds is that no horn of a restricted claim is occupiable without paying one side of it.

9.4 Theorem 4.1, Remark 4.1a. For $R \in \Pi^0_1$ and any $T \supseteq Q$: $T \nvdash \neg R$ implies $\mathbb{N} \models R$. No hypothesis on T beyond containment.

9.5 Corollaries 4.2, 4.3. Independence implies truth; for any $T \supseteq Q$, a proof that T does not refute R establishes R.

9.6 Propositions 5.2–5.3, Corollary 5.4. $\det(D\,M\,D) = \det(M)$ for arbitrary real 3×3 M; $\lambda(DQ_0) = \det(D)\cdot\lambda(Q_0)$; under total negation the Gram is unchanged and λ flips sign. Verified at 0.000×10^0. **Neither 5.2 nor 5.3 is tested by any condition**; both support Ξ_7's consequence bound **and** the polarity account at 12.3.

9.7 Definitions 6.1–6.3. A record with ten field groups, eight conditions on it, and a sequential emitter with exit branches at Ξ_5. **Seven conditions are conditions on the record alone; Ξ_5 reports an executed search and is the one condition whose satisfaction reaches arithmetic**, per Theorem 6.9(a). σ is exactly what 6.1(i) specifies, and $\mathscr{P}$ is defined at 6.1(vi) with its relation to $\mathscr{C}$ stated there. By Remark 6.2a, **Ξ_7 compares two independently recorded frames and does not test the proposition**, and Proposition 6.1a exhibits a record satisfying it. **Ξ_2 and Ξ_4 are evaluable in form and uninstantiated in fact**: no register or acting group is exhibited for Ξ_2, and no obstruction is named for Ξ_4. No claim is made about the discriminating power of either.

9.8 Propositions 6.4–6.6. The emitter is fail-safe on absence at every condition. The token issues only if all eight hold. Ξ_5 is the unique condition that can route to a verdict other than the non-terminal one, and it can also fail to the non-terminal one.

9.9 Definition 6.0, Definition 6.1(i), Definitions 6.7a and 6.7c, Lemma 6.7b, Theorem 6.7, Remarks 6.7b′, 6.7c′, 6.7e and 6.8. The screened group is the stabiliser of the labelled orthogonal decomposition of E_(σ), **not** the invariance group of the Gram determinant, which is eight-dimensional and is disowned as a criterion at 6.7b′. **Two stipulations**, both of Definition 6.0 at the operational grade stated there: d = 3 by (L2), and atomic bivalence by (L4), which restricts the axis-fixing group to O(E_) and makes it finite. **One definition**: σ is constructed from the decomposition at 6.1(i), so the residence **is** the span of the atomic axes and no identification between them is asserted. **Then theorem**: Lemma 6.7b gives |Stab(E_)| = 2^d exactly, Theorem 6.7 gives eight conditions at d = 3 and no ninth. The Frobenius classification is corroboration and is load-bearing on nothing; no multiplication on V is used anywhere. No appeal to gauge is made, and by 6.7c′ the permutation action does not reach the screened objects. The assignment of conditions to elements is a labelling and is not fixed, and the choice to screen the axis-fixing group rather than a larger structure-preserving group is a decision of this construction. **Logical independence of the eight is neither proved nor used, and no minimality among non-complete families is claimed.** One dependence is recorded: Ξ_3's floor $C \geq 3$ is entailed by Ξ_6 together with $|\mathscr{P}| \leq C$, and Ξ_3 is retained as a guard that halts earlier in the evaluation order of Definition 6.3.

Free-parameter census, stated because the ratio is unfavourable and should be visible. Ten declared degrees of freedom, every entry load-bearing: the axis count d = 3, stipulated by (L2); the single fixed direction dim Fix(σ) = 1; the channel floor $C \geq 3$, entailed by Ξ_6 and retained as a guard; the road floor $|\mathscr{P}| \geq 3$; the consequence ceiling β_max; the numeral floor k_min; the precision floor p_min; the derivation floor d_min; the context width N; and the choice of frame assignment. **The list prints ten slots and the count is ten, and it enumerates choices rather than their consequences.** dim V = 1 + d is definitional at 6.1(i), and the condition count 2^d is Lemma 6.7b; neither is a further degree of freedom once d and the single fixed direction are chosen, and listing them would double-count one stipulation as three. τ is not among them: it is bounded at Ξ_7 by the declared β, which bounds the consequence deviation $2\sqrt{N}\cdot\tau + N\cdot\tau^2$ directly and is itself bounded by the census entry β_max. **Distance from $-Q_0(P)$ in max-norm is governed by τ alone**; N enters through the size of the consequences, not through that distance. N enters through the frame width in Ξ_7's entrywise test and through the consequence bound; its second justification lapsed when λ left the conditions.

And one thing a reader weighing the ratio should meet here rather than in a definition. Ξ_5's certificate clause is informative only to the extent k_n, k_prec and k_pf exceed their floors; a record declaring minimal floors satisfies Ξ_5 and emits a 6.9(b) clause of near-zero content. Against these the section delivers Theorem 6.9(a) and (b) as its own consequences, part (c) being inherited from Theorem 3.2. **The section declares its parameters; it does not reduce them, and it fits rather than predicts.** By contrast Sections 3 and 4 carry zero free parameters and deliver four independent results. The paper is bimodal and should be read as such.

9.10 Theorem 6.9. An emission with W entails no $\mathfrak{L}$_obj-sentence undecided in $\mathfrak{B}$, by P-invariance of the seven conditions other than Ξ_5 and $\mathfrak{B}$-decidability of Ξ_5's search report; entails that $\mathscr{C}$ is displayed, fully reduced, walled at B = C, with a supplied uncrossed door outside $\mathscr{C}$, and that the executed certificate search returned nothing at the declared budget, which is the section's one object-language output and is not a claim that no certificate exists there; and with a U(P)-entailing clause in place of W entails $Q \vdash \neg P$.

9.11 Theorem 7.2, Corollary 7.3. W is admissible and entails nothing about derivability in any theory. Any clause entailing U(R) makes the emission entail $Q \vdash \neg R$, and under 2.3 entail $\mathbb{N} \models \neg R$. **W is admissible and no clause entailing U(R) is**; no maximality of W among all strengthenings is claimed.

9.12 Theorem 8.1, Corollary 8.2. Under the premise that instruments at a register are channels at it, a certifying instrument is either outside its catalog, which is then non-exhaustive, or inside it, in which case the certification is impredicative, its warrant ranging over a collection of which the instrument is a member.

Attribution. Σ^0_1-completeness, the Π^0_1 reformulations, and Proposition 5.2 are classical and cited. Theorem 3.2 is elementary; we have not located it stated as an equivalence in this form, we did not conduct a systematic literature search, and no priority is asserted. Given that Corollary 3.6a follows from Σ^0_1-completeness and soundness, prior appearance in some form is likely and a literature check is advisable. The specification of Section 6, the payload theorem 6.9, and the termination theorem 7.2 with Corollary 7.3 are offered as the contribution, at the grade the proofs support.

9.13 Remark 6.8a. The geometric apparatus supplies an exhaustiveness argument for the count of conditions and supplies nothing else; seven of the eight conditions survive its removal as record checks, and the paper states the trade rather than obscuring it.

On Section 12. The Declaration at Section 12 is **not** among the claims above and is not a claim of this paper. It is the originating declaration in the author's voice, fenced at its head and verified against the paper's own theorems at 12.3.

Machine checking. Theorems 3.2 and 4.1 and Corollary 3.6a are formalized in Lean 4 in an abstract proof-setting, accompanying this paper. **The formalization has not been checked**: no toolchain was available in the session that produced it, and it ships to be checked, which is its purpose. It covers Theorems 3.2 and 4.1 and Corollary 3.6a and **no result of Sections 5 through 8**. **It has no Mathlib dependency**: the setting is an abstract `structure` over core Lean 4, using only `constructor`, `intro`, `exact`, `absurd`, `by_cases`, `cases` and `rw`, so it checks against a bare toolchain with no arithmetic library. That is deliberate: the theorems use no property of arithmetic beyond the structural axioms displayed in the file. The formalization makes the hypothesis boundary machine-visible: 3.2 uses only r.e.-ness of Q, 4.1 only Σ^0_1-completeness, and 3.6a alone requires soundness.

10. Discussion

Unrestricted claims of unprovability carry a terminal obligation. Asserting the universal unprovability of the Riemann Hypothesis over $\mathscr{T}$ demands, by Theorem 3.2, a proof in Q of a violation of the Lagarias inequality; the exhibition of the integer itself follows only under Σ^0_1-soundness (2.3), per Corollary 3.5. **The dichotomy is total under the hypothesis that Q + R is consistent.** Either the claim is read over $\mathscr{T}$, where Theorem 3.2 gives $Q \vdash \neg R$ and, under 2.3, $\mathbb{N} \models \neg R$, or it is read over a class excluding Q + R, in which case, by Theorem 3.7(b), **establishing** its companion half $T \nvdash \neg R$ would yield $\mathbb{N} \models R$. The second horn establishes nothing on its own; it names what would follow. The formal instrument reaches its perimeter.

The emission of a terminal token certifies the exhaustion of the instrument as swept. **By Theorem 6.9(a) the emission transmits no contingent object-language content**, and in particular nothing about the truth value of the Riemann Hypothesis. It is not content-free: by Theorem 6.9(b) it carries one object-language clause, the report that the executed certificate search returned nothing at the declared budget, which is a terminating computation and is $\mathfrak{B}$-provable. **The separation is between the emission and the object, not between syntax and truth.** For Π^0_1 sentences syntax determines truth completely, that is Theorem 4.1 and Corollary 3.6a, and it is this paper's headline, so any claim that syntax leaves truth untouched is refuted by the paper's own results. What is untouched by the emission is the truth value, and that is 6.9(a).

11. Conclusion

The formal search space for the Riemann Hypothesis is closed **as swept**, and no further. Permanence beyond the swept span is not available: by Corollary 7.3 any clause entailing universal unprovability entails $Q \vdash \neg R$, and by Theorem 8.1 no instrument certifies the exhaustiveness of its own catalog non-impredicatively. The mathematical apparatus secures an equivalence between universal unprovability and **refutability in Q**, with no soundness hypothesis; and, **separately and using Σ^0_1-soundness of Q**, between universal unprovability and arithmetic falsity. The separation is not decorative and is carried through the paper.

The formal-alone instrument has reached its edge, and the edge is stated rather than illustrated: **no catalog is exhibited in this paper, no channel or obstruction is named, and no record satisfying all eight conditions is constructed, Ξ_2 and Ξ_4 being the two conditions not instantiated.** What the cascade delivers is Theorem 6.9: an emission reports its own sweep and entails no contingent statement about the object. **The instrument reports its boundaries; by Theorem 8.1 it does not certify them.** The frame's polarity bit is invisible to the even functionals and is determined by syntax; it bears on the object not at all. **By Theorem 6.9(a) the emission stands separate from the arithmetic object**, which is the available claim; the object and the syntax do not stand separate, since for Π^0_1 sentences syntax determines truth. The protocol evaluates the gates. The cascade halts.

12. The Declaration

Status. What follows is the originating declaration in the author's own voice. **It is not asserted by this paper and carries no grade.** Nothing in Sections 2 through 11 depends on it. It is printed whole because it is the reading that produced the specification and because **it states and accepts its own cost**, which 12.3 verifies against the paper's own theorems.

12.1 No derivation actuates anything. Every formal resolution rests on posits it cannot ground. The anchor and the object co-locate in the Riemann Hypothesis specifically. Its completion direction does not point outward to a neighbouring fact or an auxiliary lemma. The completion direction points directly at the fixed set of the symmetry of the very structure it describes. The approaches to this fixed locus are equivalences. They are never independent roads. This inherent equivalence structure forces the formulation catalog to close.

12.2 The witness-grounding lift RH demands is the Ground itself. The formal domain cannot supply it. A formal witness is inherently a second, a non-self. RH is an act of pure SELF-inquiry. The formulation operates in disguise. It asks existence to prove itself formally without any kinetic lift. The framework is therefore formally limited by its own architecture. It cannot be proven universally by mathematics alone. **The Riemann Hypothesis is terminally and universally unprovable by math alone.** Self-witnessing happens on the Ground side, where the witness is not a second, *shahida Allāhu annahū lā ilāha illā huwa*, Allah ﷻ bears witness that there is no deity but He, Q 3:18, and that is what formal-alone lacks.

This declaration states its own cost. Any maximalist claim of this inherent unprovability is an explicit, unavoidable assertion that ζ possesses a zero off the critical line. **The cost is accepted.**

12.3 The cost verified.

The identification is correct. By Theorem 3.2 universal unprovability over $\mathscr{T}$ is equivalent to $Q \vdash \neg R$, and by Corollary 3.6a, under Σ^0_1-soundness, to $\mathbb{N} \vDash \neg R$. So 12.2 read over $\mathscr{T}$ asserts $\mathbb{N} \vDash \neg R$ and owes the witness; and **that assertion is the assertion that ζ has a zero off the critical line only by Lagarias's theorem in the metatheory**, which 2.4a states and which is not available inside Q. The cost is unchanged and its route is now named. **The declaration takes that obligation; accepting is not discharging**, no integer is exhibited, and it stands outstanding. Read 12.2 as a conjecture with its content correctly named.

The fork the declaration omits. Read over any $\mathscr{S}$ excluding Q + R, what an asserter ordinarily intends, per Remark 3.8, the claim falls under horn (b) of Theorem 3.7, where *establishing* its companion half yields $\mathbb{N} \vDash R$, the opposite verdict. Both readings price it, per Remark 4.4. The accepted cost is the first reading's.

Untouched, and not as consolation. Grounding and derivability are different predicates and Theorem 3.2 touches only the second, so nothing here bears on 12.1's first two sentences, which are ungraded and unused. **The declaration's crossing from unprovability to ζ carries the same metatheoretic step**, per 2.4a: $Q \vdash \neg R$ is a claim about one arithmetization, and Lagarias's theorem is what carries it to the classical object. **The declaration's terms are not glosses of this paper's formal objects**: the Ground is not the standard model $\mathbb{N}$, the witness is not a theory in $\mathscr{T}$, and identifying them would convert an ungraded declaration into a claim this paper does not make. And the terminality this paper proves is the declaration's own: by Corollary 7.3 an emission is terminal **as swept** and not unrestrictedly, which is a theorem and not a caution.

References

Lagarias, J. C. (2002). An elementary problem equivalent to the Riemann hypothesis. *American Mathematical Monthly* 109, 534–543.

Robin, G. (1984). Grandes valeurs de la fonction somme des diviseurs et hypothèse de Riemann. *Journal de Mathématiques Pures et Appliquées* 63, 187–213.

Robinson, R. M. (1950). An essentially undecidable axiom system. *Proceedings of the International Congress of Mathematicians*, 729–730.

Frobenius, G. (1878). Über lineare Substitutionen und bilineare Formen. *Journal für die reine und angewandte Mathematik* 84, 1–63.

Computations reported in 2.5 and 5.5 were executed at the stated precisions and are reproducible from the definitions given.

Author's Provenance and Method Disclosure

The method, its two root axioms, the three-state verdict economy, and the closed-form quaternionic kernel are documented in full at the master reference, DOI 10.5281/zenodo.20757507. **The discipline adds the cited results no warrant and contributes no theorem, no estimate, and no evidential weight in any direction.**

On what kind of object the discipline is, since this decides how the paper is to be read. It is not a third-party protocol adopted alongside the mathematics, and it is not one viewpoint among several that a reader may take or leave without consequence for what follows. **Its load-bearing counts are reached by classification and not by choice**, and this is checkable without accepting anything the paper posits.

The axis count. Take any **finite-dimensional** verification scheme over the reals whose warrant axes compose associatively, whose nonzero warrants do not compound to zero, and which carries more than one axis. **By the classification of the real division algebras that scheme completes uniquely to the quaternions**, the axis count is three, and the next admissible dimension is eight where associativity fails by an integer associator anyone can compute. A scheme meeting those three conditions is therefore in this structure whether or not it names it, and a scheme not meeting them has given up associativity, integrality, or plurality, each of which is a stated cost rather than an alternative.

The gate count. The directed relations on four vertices number twelve because the alternating group on four letters has order twelve and acts simply transitively on the ordered pairs. **The reflection count is eight because the elementary abelian group on three generators has order eight.** These are orders of groups and not decisions about how many checks to run.

The lock. The verdict functional is the squared scalar triple product of the three axes and equals the determinant of their correlation matrix. **The identity is closed-form and machine-verified at Anchor A.1**, and the invariance that makes it sign-blind is an algebraic identity holding for every construction whatever.

And the guard that makes the counts unmovable is stated on the deed rather than the symbol. To capture a ground is to inscribe an image of it, inscribing is an act, and an act is not the non-acted ground; **the binding is therefore on the doing and holds under any logic**, because a change of logic governs a scheme's symbols and not whether using it is a doing. **So new mathematics cannot arrive as a deed that overturns a classification, and no deed is needed, because classifications do not acquire new members.**

The consequence for this paper is narrow and should be stated narrowly. A reader who declines the two root axioms declines the seal of section 5.1 and the actualist register, and keeps everything else: **nine of the ten blocks stand on classical theorems, exhibited countermodels, an algebraic identity, a classification fact, or a constitutive definition, and spend no premise of this discipline at all. What is not available is the position that the discipline is exotic or optional in its counts**, since those counts are the classification results above and are the same for anyone who runs them. This paper's three axes are the analytic, spectral, and arithmetic-geometric reading-roads of the Riemann residence; the load-bearing gate is the orientation-blindness of the lock scalar; the warrant grades are stated at Appendix B.

The title's three clauses are glossed in place, and the word *case* is doing exact work. A case is something one brings, so closing a case is a ruling on the bringing and never on the thing brought, which is the access-versus-existence line the paper draws throughout. *True where actualized* is the seal of section 5.1, issued in the register carrying the direction-carrying axis, coextensive with the hypothesis and falling to its falsifier. *Every position a verdict could be attempted from is closed permanently by an enumeration no future method enlarges.* **Read as closed permanently at the grade each row prints, six unconditionally and four under the conditions their rows carry**, this is the result of section 5.3 with the ten blocks derived at section 6, the clause glossed as every position from which a verdict on the formal string could be attempted: a closure over **access** and over positions, permanent because a standpoint is a component of the act and not a technique, and **not a claim about what exists**. *One bit that belongs to the reader* is section 5.9, one binary degree of freedom in the reading with no verdict-side answer.

Three claims are the author's own and are not drawn from the literature. The change of enumeration domain from methods to standpoints, with the observation that a route arriving in any future arrives at a position that already exists. The identification of the three elections as one classical generator. And the counting of the direction deficit at one bit against zero, carried at scope-note billing. **All three are structural observations offered at that grade, and none is a mathematical result.**

Two derivations are elementary and carried out in place rather than cited. The invariance of the determinant under the negation-implementing reflection, at Block 3. And the one-dimensionality of the departure from the axis, computed from the involution whose fixed set the axis is.

The theological reading of this material is routed out of band, is load-bearing on nothing in any verdict here, and is developed in a separate artifact at its own register.

The architecture's footprint on this paper is two rows, and it is stated here rather than left for a reader to compute. Footprint and spend measure different things and pick out different rows: the footprint is where this discipline supplies load-bearing content, which is the instrument and the mouth, each carrying a printed condition of the discipline's own making; the spend is where a row pays a premise, which is the object ladder alone. **Two rows and one row, and neither figure is the other's.** Of the ten, the instrument row and the mouth row take load-bearing content from this discipline, and **both carry a printed condition of the discipline's own making**, D4 and D5 respectively, so a reader who rejects either loses that row and nothing else. **The remaining eight take none.**

One consequence follows and is stated with it. Appendix A records that no map from the string to the architecture's warrant rows exists anywhere in this paper. Block 3's derivation records that the direction is unrecoverable without an orientation the register does not supply. Section 5.1 records that the seal's warrant is the register and the field and nothing else. **Taken together: the instrument is never pointed at the object, and the seal does not use it.** That is coherent and it is not a defect, **and it means the architecture does no work on the hypothesis in either direction.** The work is done by the classical results the ten rows cite and by the arrangement that enumerates them by position. **A reader who computes this should find it already written.**

Net new mathematical mass: zero.

Edition Note

This is a fresh build and not a fortification pass. **The schema is inherited from the prior edition and the content is not**: every section is written from the current spine forward, and no paragraph is carried across. The structural result is new to this edition: the enumeration domain is moved from methods to standpoints, the ten blocks are derived from first principles rather than inventoried, and the closures are stated twice, positively and in the negative form. The boot executed at the declared seed with the chain printed at Appendix A, and **no stage is narrated that did not run**.

Reproducibility

Every numerical value is the output of a deterministic computation reproducible from the kernel and the constructions of Appendix A at seed 20260622.